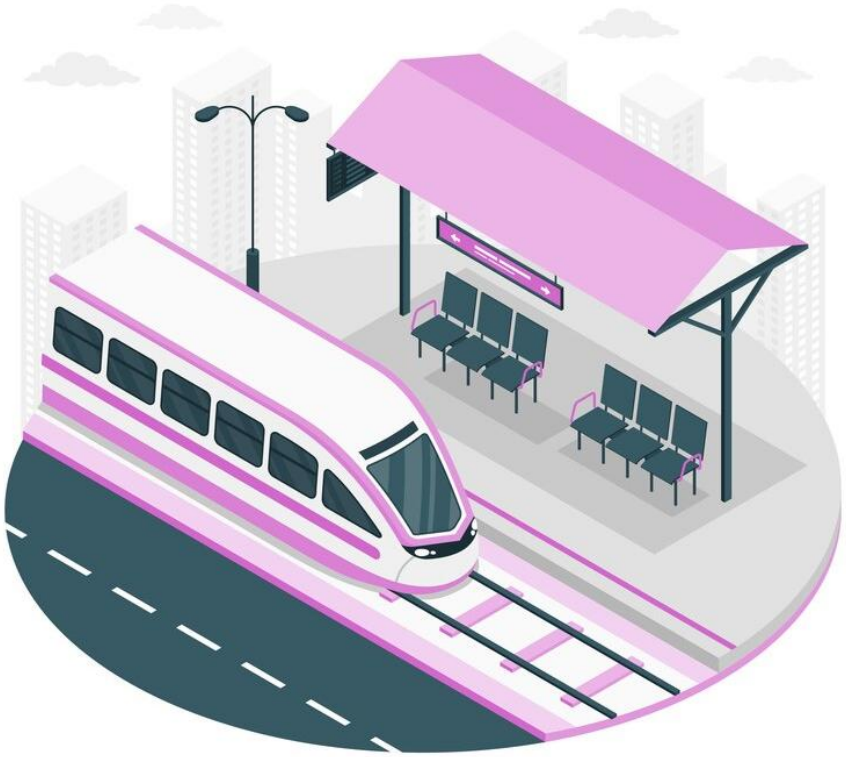
CIS 321 - Object Oriented Programming 2

Group No. 6
Team No. 1
Ms. Dalal Aldowaihi

TrainX Station System



Role	Student Name	Student ID
Leader	Imtinan Bani Alhajri	2230004226
Member	Shahad Mohammed Alqarni	2230003336
Member	Zahra Redha Al-Thunayan	2230004315
Member	Amna Hani Fadel	2230002174
Member	Bayan Ahmed Alghamdi	2230001087
Member	Renad Abdulghani Mengash	2230003701

Description:

TrainX is a railway app designed to enhance the experiences of passengers, employees, and management. With its user-friendly interface and modern features, our app provides a streamlined platform for providing services to passengers. Passengers can easily purchase tickets and pay through the app or later via cash. They can also view their tickets and update or delete train status information. **TrainX** also supports railway operations by managing employee information, allowing managers to edit and update data as needed, and empowering them to add, delete, or modify train data.

Logical analysis & Business:

Passengers and managers have different levels of access to the system. Initially, managers have full access to the system, and employees can modify the services provided to customers. Passengers only have access to their own reservations, which they can modify, delete, or book other trips. They cannot access administrative services.

Difficulties & How Problems Were Solved:

1. Distributing work among members: It was difficult for us to distribute tasks. We were open and honest about everyone's abilities, interests, and accessibility, which allowed us to overcome this. Furthermore, we created a shared folder using OneDrive to monitor assignments and their due dates. For all previous submissions, we relied on Google Docs so that each student could write their part at their own convenient time, which helped provide greater work freedom.

2. We encountered communication issues. We established proper communication channels and rules for addressing the issue. We held several online meetings to work together on linking our individual parts and to address any issues that arose together.

3. Errors were persistent problems throughout our project. One such issue was the failure to delete information from the database when the application was being deleted. We addressed this issue in an online meeting. We encountered several other issues, and these were divided among us to address them, such as printing invoices and passing prices on multiple interfaces. We also addressed the design together in another meeting.

4. Debugging:

During the development and integration of the system, we encountered a wide range of technical challenges. As the project was built collaboratively by multiple team members, integrating the various components into a single functional system led to several issues. These included common and critical exceptions such as:

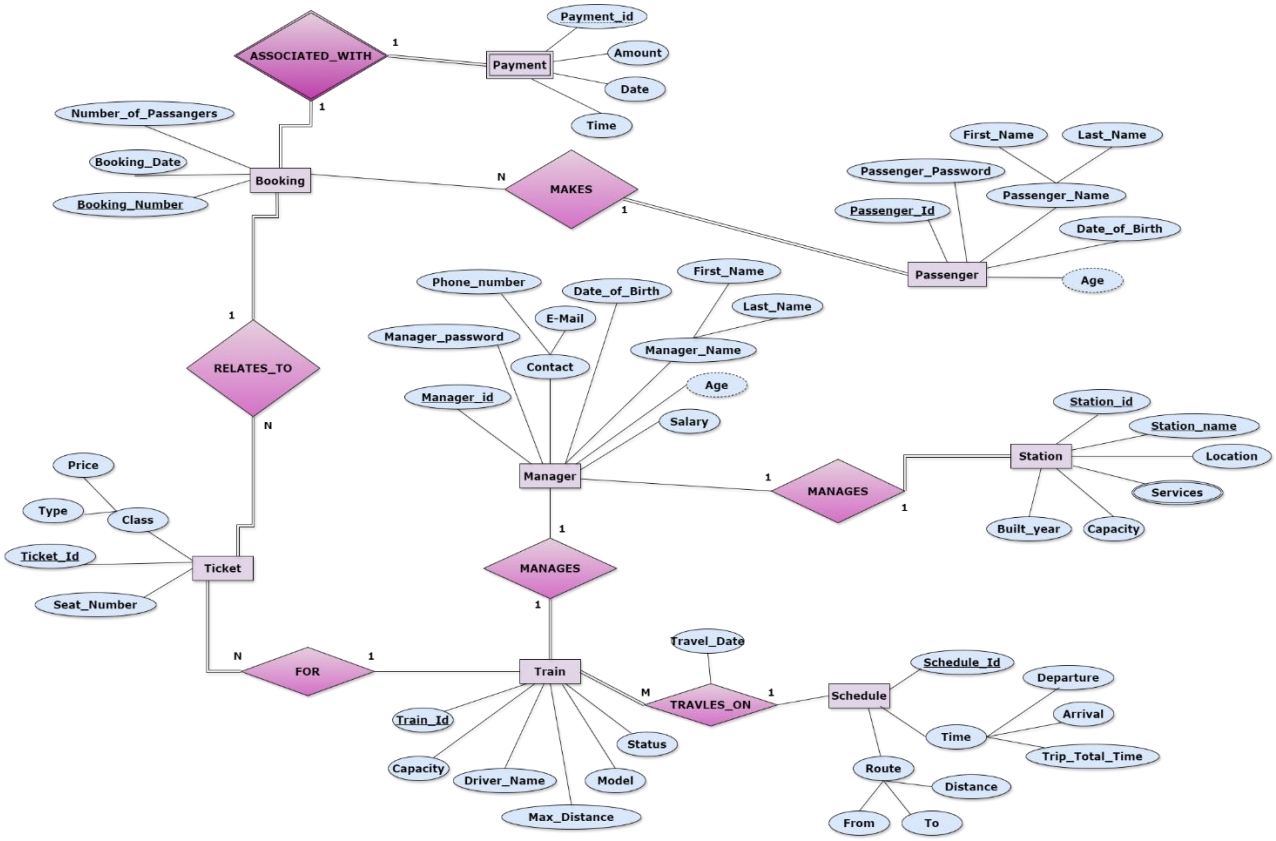
- NullPointerException
- IndexOutOfBoundsException
- SQLSyntaxErrorException
- java.lang.RuntimeException: Uncompilable source code
- NoSuchMethodException
- Issues with database connections being null or unexpectedly closed

The Uncompilable source code exception often occurred due to unresolved references, missing or mismatched method calls, and code that failed to compile due to syntax or logic issues introduced during integration. The NoSuchMethodException typically arose when trying to call a method that either didn't exist or wasn't properly declared, often due to miscommunication between different modules or incorrect method signatures.

To address these challenges, a dedicated debugging team was formed, led by our team leader, Imtinan. The team worked efficiently to trace each problem, diagnose its root cause, and apply suitable fixes. Their teamwork and attention to detail were vital in ensuring program stability and preserving data consistency across the system. Through their consistent effort and collaboration, the debugging team successfully resolved all major errors, allowing the system to function smoothly and reliably by the end of development.

Our Logo:

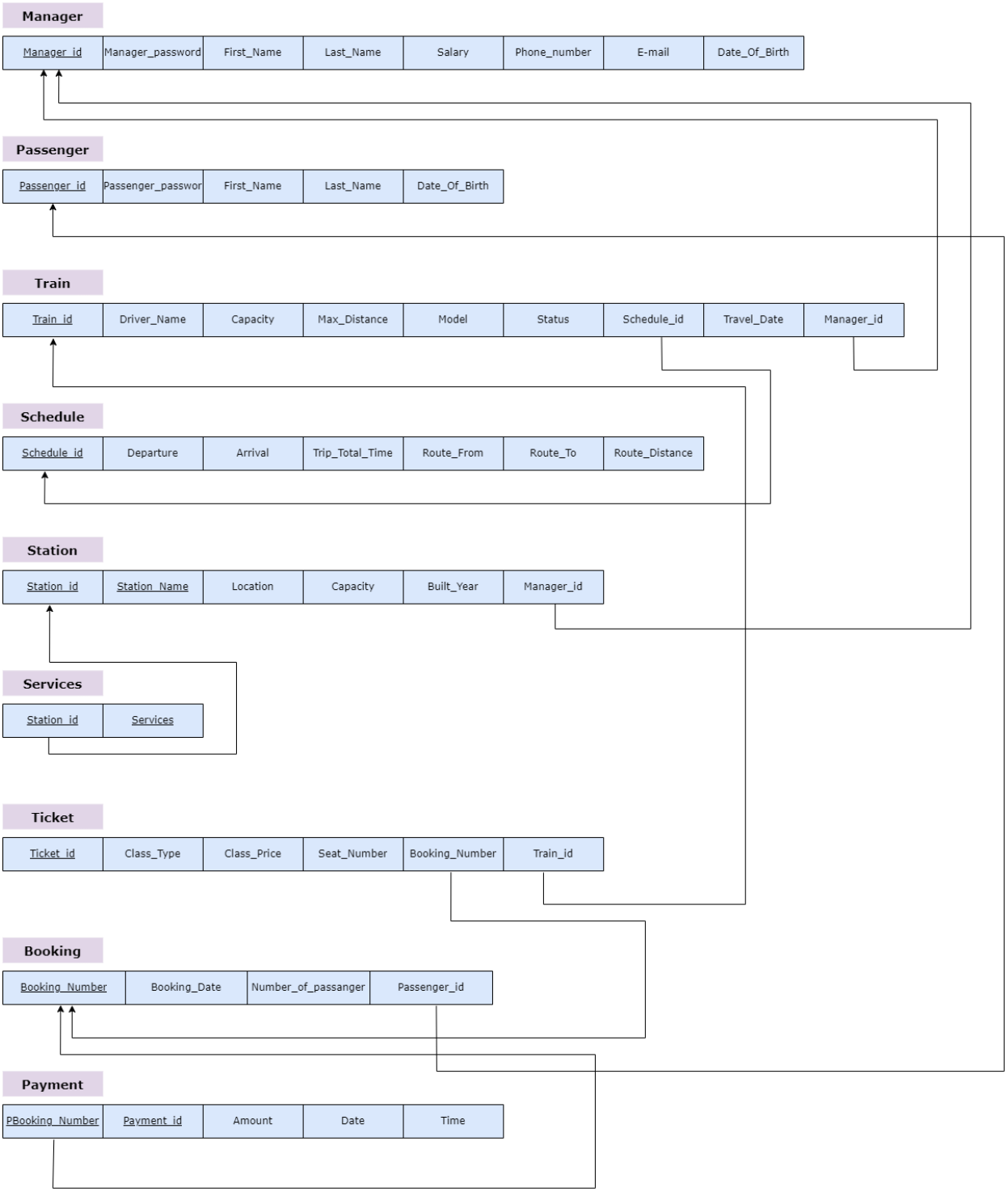
ER Diagram:



The ER diagram represents a train railway system, composed of the following entities:

Passengers	Managers	Trains	Schedule
Stations	Tickets	Booking	Payment

Database Schema:



Checklist of Requirements Fulfilled in the Project:

Function / Application Aspect	Classes Management	Data Export/Import	Professional Appearance	Error Handling	Data Access Tools and Techniques
User's Log-in (different types of users :Managers & passengers)			✓	✓	✓
Managers (add\modify\delete a trip) (add\delete\update train)			✓	✓	✓
Passenger's (book\update\delete a trip\payment)			✓	✓	✓
Request tickets prices			✓		✓
User's Statistics	✓	✓			
Trips statistics	✓	✓			
export the seats and prices for class service		✓		✓	

Tables Filled with data:

Manager

Manager_Id	Manager_Password	First_Name	Last_Name	Email	Phone_Number	Birth_Date	Salary
M001	AmFa2174	Amna	Fadel	Amna@email.com	+966556954366	2005-03-14	60000.00
M002	ReMe3701	Renad	Mengash	Renad@email.com	+966587654321	2001-06-25	68000.00
M003	ImAl4226	Imtinan	Alhajri	Imtinan@email.com	+966551686144	2004-04-02	80000.00
M004	BaAl1087	Bayan	Alghamdi	Bayan@email.com	+966551112233	1998-09-30	62000.00
M005	ShAl3336	Shahad	Alqarni	Shahad@email.com	+966554433221	2002-12-12	59000.00
M006	ZaAl4315	Zahra	Al-thunayan	Zahra@email.com	+966545455944	2004-07-03	70000.00
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Passenger

Passenger_Id	Passenger_Pass	First_Name	Last_Name	PassengerDOB
P001	HaRe1976	Hawraa	Redha	1976-10-18
P002	MaAl2004	Maha	AlQahtani	2005-02-10
P003	MaHa2004	Maria	Hadi	2004-03-30
P004	MoAb2001	Mohammad	Abdullah	2001-09-01
P005	LaKh1998	Layla	Khalid	1998-07-22
P006	YoSa1995	Yousef	Salem	1995-12-05
P007	HaRa2000	Hana	Rashid	2000-04-15
P008	SaHa1996	Sami	Hassan	1996-03-10
P009	NaMi1993	Nadia	Mansoor	1993-11-30
P010	FaAl1997	Faisal	Ali	1997-08-20
NULL	NULL	NULL	NULL	NULL

Trains

Train_Id	Driver_Name	Capacity	Max_Distance	Model	Train_Status	Manager_Id
1	Lina Abdullah	28	1500.00	Model 2023	Active	M005
2	Maha Ali	20	1800.00	Model 2022	Active	M001
3	Shahad Mohammed	25	1200.00	Model 2021	Active	M006
4	Rana Ahmed	24	2000.00	Model 2024	Delayed	M003
5	Layla Hassan	22	1300.00	Model 2020	Active	M006
6	Nour Khalid	27	1600.00	Model 2018	Active	M002
7	Omar Farid	28	1800.00	Model 2022	Active	M004
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Schedule

Schedule_id	Departure	Arrival	Trip_Total_Time	Route_From	Route_To	Route_Distance	Train_Id
101	2025-05-02	2025-05-02	7 Hours	Riyadh	Jeddah	955	1
102	2025-05-02	2025-05-02	3 Hours	Riyadh	Dammam	410	2
103	2025-05-02	2025-05-02	3 Hours	Medina	Jeddah	408	3
104	2025-05-02	2025-05-02	3 Hours	Medina	Jeddah	408	4
105	2025-05-02	2025-05-02	7 Hours	Riyadh	Jeddah	955	5
106	2025-05-02	2025-05-02	10 Hours	Jeddah	Dammam	1362	6
107	2025-05-02	2025-05-02	9 Hours	Medina	Dammam	1226	7
108	2025-05-03	2025-05-03	10 Hours	Jeddah	Dammam	1362	1
109	2025-05-03	2025-05-03	6 Hours	Riyadh	Medina	828	2
110	2025-05-03	2025-05-03	3 Hours	Dammam	Riyadh	410	3
111	2025-05-03	2025-05-03	3 Hours	Jeddah	Medina	408	4
112	2025-05-03	2025-05-03	6 Hours	Medina	Riyadh	828	5
113	2025-05-03	2025-05-03	3 Hours	Dammam	Riyadh	410	6
114	2025-05-03	2025-05-03	3 Hours	Dammam	Riyadh	410	7
115	2025-05-04	2025-05-04	3 Hours	Medina	Jeddah	408	1
116	2025-05-04	2025-05-04	6 Hours	Riyadh	Medina	828	2
117	2025-05-04	2025-05-04	7 Hours	Riyadh	Jeddah	955	3
118	2025-05-04	2025-05-04	3 Hours	Medina	Jeddah	408	4
119	2025-05-04	2025-05-04	7 Hours	Jeddah	Riyadh	955	5
120	2025-05-04	2025-05-04	7 Hours	Jeddah	Riyadh	955	6
121	2025-05-04	2025-05-04	7 Hours	Jeddah	Riyadh	955	7
122	2025-05-05	2025-05-05	3 Hours	Riyadh	Dammam	410	1
123	2025-05-05	2025-05-05	9 Hours	Medina	Dammam	1226	2
124	2025-05-05	2025-05-05	10 Hours	Dammam	Jeddah	1362	3
125	2025-05-05	2025-05-05	9 Hours	Medina	Dammam	1226	4
126	2025-05-05	2025-05-05	3 Hours	Riyadh	Dammam	410	5
127	2025-05-05	2025-05-05	9 Hours	Dammam	Medina	1226	6
128	2025-05-05	2025-05-05	6 Hours	Riyadh	Medina	828	7
129	2025-05-06	2025-05-06	3 Hours	Jeddah	Medina	408	1
130	2025-05-06	2025-05-06	3 Hours	Medina	Jeddah	408	2
131	2025-05-06	2025-05-06	10 Hours	Jeddah	Dammam	1362	3
132	2025-05-06	2025-05-06	10 Hours	Jeddah	Dammam	1362	4
133	2025-05-06	2025-05-06	9 Hours	Dammam	Medina	1226	5
134	2025-05-06	2025-05-06	3 Hours	Dammam	Riyadh	410	6
135	2025-05-06	2025-05-06	9 Hours	Dammam	Medina	1226	7
136	2025-05-07	2025-05-07	3 Hours	Medina	Jeddah	408	1

Station

Station_Id	Station_Name	Location	Capacity	Built_Year	Manager_Id
1	Riyadh Station	Riyadh	1500	2010-07-24	M001
2	Jeddah Station	Jeddah	1400	2015-09-22	M002
3	Dammam Station	Dammam	1000	2023-01-01	M003
4	Almadinah Station	Almadinah	1200	2001-01-01	M004
NULL	NULL	NULL	NULL	NULL	NULL

Service

Station_Id	Service
1	Bathroom
1	Coffee shop
1	Parking
1	Restaurant
1	Waiting room
2	Bathroom
2	Coffee shop
2	Parking
2	Restaurant
2	Waiting room
3	Bathroom
3	Coffee shop
3	Parking
3	Restaurant
3	Waiting room
4	Bathroom
4	Coffee shop
4	Parking
4	Restaurant

Tickets

Ticket_Id	Class_Type	Class_Price	Seat_Number	Booking_Numb...	Train_Id	Travel_date
TK001	First Class	250.00	A1	BK001	1	2025-05-02
TK002	Economy	150.00	A3	BK002	2	2025-05-02
TK003	Economy	150.00	B3	BK002	2	2025-05-02
TK004	First Class	250.00	C1	BK003	5	2025-05-02
TK005	Economy	150.00	B4	BK004	4	2025-05-02
TK006	Economy	150.00	C4	BK004	4	2025-05-02
TK007	Economy	150.00	D4	BK004	4	2025-05-02
TK008	First Class	250.00	A1	BK005	5	2025-05-02
TK009	First Class	250.00	B1	BK005	5	2025-05-02
TK010	Economy	150.00	A3	BK006	6	2025-05-02
TK011	First Class	250.00	B1	BK007	7	2025-05-02
TK012	First Class	250.00	A1	BK007	7	2025-05-02

Booking

Booking_Numb...	Booking_Date	Number_of_Passenge...	Passenger_Id
BK001	2025-01-01	1	P001
BK002	2025-02-09	2	P002
BK003	2025-02-23	1	P003
BK004	2025-03-04	3	P004
BK005	2025-04-05	2	P005
BK006	2025-04-18	1	P006
BK007	2025-03-01	2	P007
BK008	2025-02-15	3	P008
BK009	2025-04-10	1	P009
BK010	2025-01-20	3	P010
BK011	2025-03-05	2	P001
BK012	2025-04-15	1	P002
BK013	2025-03-28	2	P004
BK014	2025-02-15	3	P004
BK015	2025-04-19	1	P009
BK016	2025-03-20	3	P010
BK017	2025-01-05	1	P004
BK018	2025-02-20	1	P005
NULL	NULL	NULL	NULL

Payment

Payment_id	Amount	Payment_Date	Payment_Time	Booking_Numb...
1	250.00	2025-01-01	13:00:00	BK001
2	300.00	2025-02-09	14:30:00	BK002
3	250.00	2025-02-23	11:15:00	BK003
4	750.00	2025-03-04	10:45:00	BK004
5	500.00	2025-04-05	15:20:00	BK005
6	150.00	2025-04-18	16:30:00	BK006
7	500.00	2025-03-01	12:00:00	BK007
8	750.00	2025-02-15	09:00:00	BK008
9	250.00	2025-04-10	14:00:00	BK009
10	750.00	2025-01-20	17:00:00	BK010
11	500.00	2025-03-05	11:00:00	BK011
12	150.00	2025-04-15	13:30:00	BK012
13	300.00	2025-03-28	15:45:00	BK013
14	750.00	2025-02-15	10:30:00	BK014
15	250.00	2025-04-19	09:15:00	BK015
16	750.00	2025-03-20	16:00:00	BK016
17	150.00	2025-01-05	12:30:00	BK017
18	150.00	2025-02-20	14:45:00	BK018
NULL	NULL	NULL	NULL	NULL

System Users:

No.	User	Privileges and Roles
1	Manager	Managers can register through the app and manage several different things: trains, and employees. They can add new trains, edit their data, or delete them. And they can view all the booking details.
2	Passenger	Passenger can register through the app and has four options: book a new trip, modify previous booking, cancel a booking or to view his tickets.

How Were the Tasks Distributed Equally Between Members?

The tasks distribution between members are as follows:

No.	Name	Tables	Interfaces
1.	Zahra	Passenger	PassModifySeats, PassModifyTrip, PassViewTicket
2.	Renad	Ticket	Payment, View_all_Trains, Manage_Employee, Update_Manage
3.	Imtinan	Booking, payment	MnaagerViewBooking, PassDeleteTicket, PassMainPage
4.	Bayan	Schedule, Station, Services	TravelDetails, ConfirmYourTripDetails
5.	Amna	Manager	Home page, LogInManager, UserLogin, Signup
6.	Shahad	Train	ManagerControlCenter, AddTrain, UpdateTrain

More details on tasks distribution:

- Zahra: **PassModifyTrip** interface allows the passenger to view their current bookings through a combo box, when a booking is selected for modification, the details of the existing trip are displayed in a table for reference. After selecting a booking, they can choose new travel details, including the departure date, origin, and destination. Based on the new input, matching available trips are displayed. The user has the option of selecting a one-way trip or a round trip using a radio button. If a round trip is selected, an additional date chooser appears (JDateChooser) for selecting the return date, and corresponding return trips are shown.

To ensure friendly platform for each one of the following possible errors, a message is displayed explaining what should the passenger do:

Error Handling:

- The passengers are not allowed to select the same city for both departure and destination.
- Selecting a travel date in the past or a return date earlier than the departure date is prevented.
- The passenger cannot proceed to the next page unless all required fields are completed: travel date, departure and destination cities, trip type (one-way or round-trip), and a selected trip from the displayed schedule.
- If any mandatory information is missing, or a trip from the available options is not selected, the system blocks navigation to the next GUI.

PassModifySeats interface shown. After updating the trip information in the previous screen, the passenger is directed to the Select Your Trip Details interface. This GUI contains a panel that visually represents the train layout, including available seats. There are three seat categories: First Class (displayed in blue), Economy (displayed in red), and Unavailable (displayed in black). When a passenger selects a seat, it displays a white seat to indicate it has been chosen.

The interface includes two text fields: one displays the currently selected seat, and the other dynamically calculates and shows the total fare based on the selected seat(s). Additionally, a text area provides a summary that includes both the old and updated booking information, which is refreshed in real-time as seats are selected or released.

Pay Modification Fee GUI, after selecting a seat, the passenger is taken to the Pay Modification Fee interface. This screen displays a breakdown of the booking costs: the original booking fee and the updated booking fee, which includes the new ticket price along with a 50 Riyal modification fee. Based on the fare difference, one of the three actions can occur:

1. No Additional Payment Required: If there is no fee to pay, the passenger can simply click the Confirm Update button to proceed.

2. Additional Payment Required: If the updated trip costs more, the passenger clicks the Pay button. The program then updates the database by deleting the old tickets and generating new ones based on the latest selections. The passenger is instructed to visit the nearest station to complete the payment.

3. Refund Due: If the updated booking costs less, the passenger clicks the Refund button. The program deletes the existing tickets, generates new ones, and directs the passenger to the nearest station to receive the applicable refund.

- Renad: The **Manage_Employee** interface is a management tool designed specifically for handling employee data from a database. It loads and displays all employee records in a structured table format, making it easy for the manager to view important details such as employee names, contact information, and other relevant fields. One of its key features is search functionality, which allows the manager to quickly filter and locate employees by entering a first name. This helps save time and improves efficiency, especially when dealing with many records.

  In addition, the interface includes an update feature, when the manager selects an employee from the table, they can open an update window or page to edit the employee's details. After making changes, the system automatically returns the user to the main management page, where they can continue working without losing their place.

  The **View_all_Trains** class is part of the manager interface and is designed to display a complete list of all trains in the system in a well-organized table. When the manager opens this interface, it automatically connects to the database, fetches all train details, and presents them in a clear table view showing the train ID, driver name, capacity, maximum distance, model, and status. The table is formatted for easy readability, helping the manager quickly scan and review important operational details.

  The **Payment** interface is designed to manage the payment process in a flexible and user-friendly way. It offers two different payment methods, card payment and cash payment. Once the user selects their preferred method.

  After the user selects a payment using card and payment is successfully processed, the interface will show the user the print bill button. This invoice contains important details about the transaction. The invoice is then saved as a text file on the desktop.

- Imtinan: She took the initiative to be our leader, organized the workflow, opened meetings to effectively communicate between group members in which she cleared the issues regarding the program, guided group members, debugged the programs clearing it from errors establishing dynamic flow of the interfaces.

  PassMainPage

  The PassMainPage serves as the central navigation point for passengers after logging into the TrainX Railway system. It greets the user with a personalized welcome message and presents four key action buttons: "New Trip," "Modify Trip," "Delete Trip," and "View My Ticket."

These buttons provide direct access to the core functionalities of the system, allowing passengers to book new journeys, modify existing reservations, cancel unwanted trips, and view their ticket details. The interface also prominently displays the TrainX Railway logo, reinforcing brand identity, and includes a "Return" button for easy navigation back to previous screens or the login page.

Delete Your Trip GUI

The Delete Your Trip interface is accessed through the PassMainPage GUI by clicking the Delete Trip button. This screen displays a table listing all tickets currently booked by the passenger. The user can select one or multiple tickets to delete. When the Delete My Trip button is clicked, a confirmation dialog appears to verify the passenger's decision. If the passenger confirms the chosen tickets are removed from the database. The table is then updated instantly to reflect the changes in real time.

Booking Statistics GUI

The Booking Statistics interface is accessed from the Manager Control Center by clicking the View Booking Statistics button. This GUI displays a table at its center showing all bookings made in the train system. It includes a month chooser component that allows the manager to filter and view booking statistics for a specific month. By default, the table shows all existing bookings. Once a month is selected, the statistics update dynamically to reflect the data for that month. Additionally, a Return to Manager Control Center button allows the manager to easily navigate back to the main control panel.

- **Bayan**: At first, she designed the **TravelDetails** interface for the traveler to specify the information about the trip he wants to book. She created a panel divided by Grid layouts into buttons representing seats, whether economy class or first class, and She placed illustrative images below it to clarify the image of each class. If there is a trip with that information the seat panel will enable the user to choose a seat, When the user clicks on the seat he wants to book, its color changes to specify it. On the panel on the right, the rest of the trip information is specified using the calendar. The user chooses the departure date of the trip (he can enter the date manually if he writes it in the correct format), then chooses the trip path from and to any city and the type of trip (round trip, one-way). If the user chooses a round trip path, the return trip date will be shown to the user. Also, the user will not be able to choose a return date before the departure date. When the user clicks confirm, the program will verify that all the information is filled in, otherwise he will not be able to move.

  As for the next interface **ConfirmYourTripDetails** it displays the data that the user entered previously before transferring him to the payment page so that he can confirm the entered

information and choose it. She created a method that retrieves the chairs that the user chose and selects them in the confirmation interface.

She also wrote a method responsible for saving a train reservation invoice in text (txt) format to the desktop, including the reservation details and selected seats. A random 7-digit reservation number is generated, and a path is created to save the invoice on the desktop, named TrainX_Invoice_<InvoiceNo>.txt. If the desktop doesn't exist (rarely), it is created. The method starts by opening the file and writing to it using UTF-8 encoding. Then, the data is written: the invoice title (TrainX) and reservation number, train line information, seats and their pricesss, the total price (totalPriceField.getText()), travel details (date, from city, to city, and return date, if applicable), and finally, the thank-you note.

She was also in charge of the files, always designing them, organizing them, putting the main headings in them, checking the margins, rules and wording, organizing and arranging the colors and images of the file and filling the milestone 3 requirement.

- **Amna**: In the homepage, users have the option to either log in or sign up. There's also an option to exit the application if they don't want to continue. From this page, a user can choose to log in either as a manager or as a Passenger. Depending on what they select, the application routes them to the appropriate login process. If the user already has an account, they can proceed to the login. If not, they can go to the sign-up page and create one.

  When logging in as a manager, the user is required to enter their email and password. The system then checks these credentials against the database. If the login is successful, the user is redirected to the Manager Control Center, where they can manage the system, access data, and perform administrative tasks.

  For Passengers, the login process is slightly different. They must provide their first name, last name, and password. These details are validated with the database. If correct, the user is taken to the Passenger Dashboard, where they can interact with the user-facing features of the system.

  If the user doesn't have an account, they can go to the sign-up page. Here, they are asked to provide their first name, last name, password, and birthdate. The system checks to make sure the user is at least 18 years old before allowing registration. The password must follow a specific policy, it must include at least three letters, three digits, and may also contain an underscore (_) or dot (.), though those are optional.

  Once the sign-up form is successfully submitted and all validations are passed, the new user is added to the database. They can then log in using the credentials they just created.

Overall, the system handles two types of users with separate login paths, including validation for secure and proper data handling, and provides access to different areas of the application based on the user's role.

- **Shahad**: She design the **ManagerControlCenter** interface, it'll be the interface where managers can do the action provided for them which are "Add Train", "Update", "View All Trains", "Manage Employees", View Bookings Statistics", the interface has button for each action which will take the manager to the selected action interface.

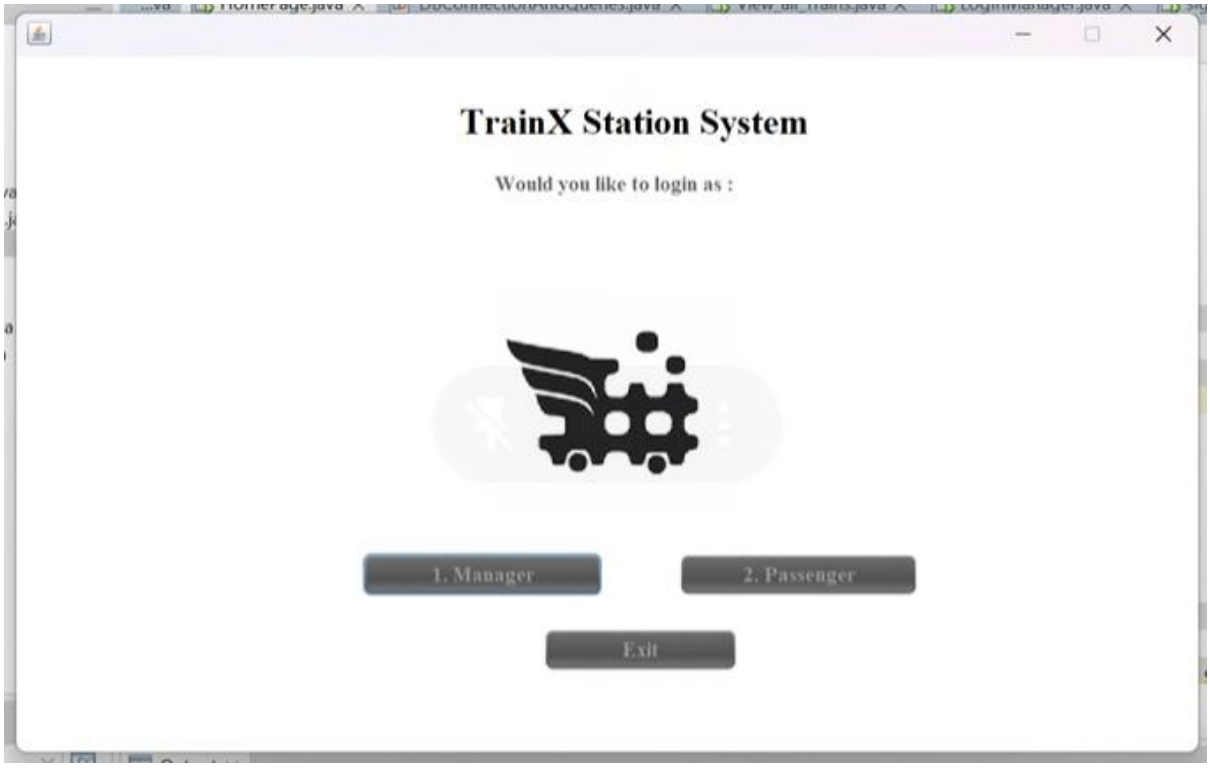
  In the **AddTrain** interface it'll be the interface where managers will add a new train to our system and it has all needed details about the train like "train id", "model", "capacity", driver name", "status", "manager id", "maximum distance" and all these fields are required except for the status and manager id since we have a default value for each. Also, we must check some validations for the details for example "train id should be a unique id, so no duplicated train ids will be allowed", and "capacity should be between 1 - 28" and more. After filling in the textfields manager will be able to add the train to the system successfully. Also, the manager could cancel adding the train and return to the manager's control center.

  In the **UpdateTrain** interface it'll be the interface where managers will be able to update an existing train, first manager will choose the train id to be updated, then click "load details" which will get the details of the specified train and will show it on the textfileds, the only data could be updated for the train are "driver name", "capacity", "manager id", and "status" while the rest of the fields will be uneditable. The same required validations in the add train interface will be applied in the update train interface. After filling the editable fields, managers will click "update", and the train will be updated in our system successfully. Also, the manager could cancel and return to the manager's control center.

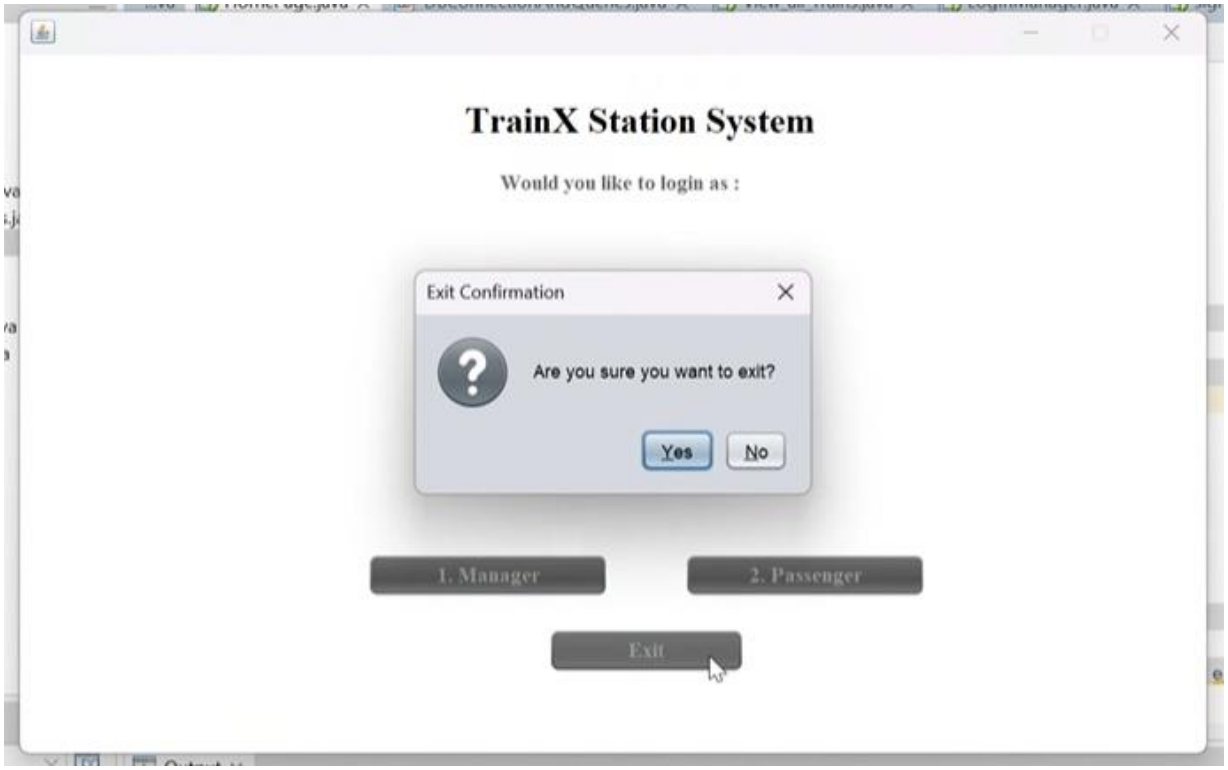
General comments:

While creating our TrainX Railway System, we learned a great deal and became even more passionate about the science and technology behind object-oriented programming. This was a rewarding experience, and we hope that the results of our work meet all our needs.

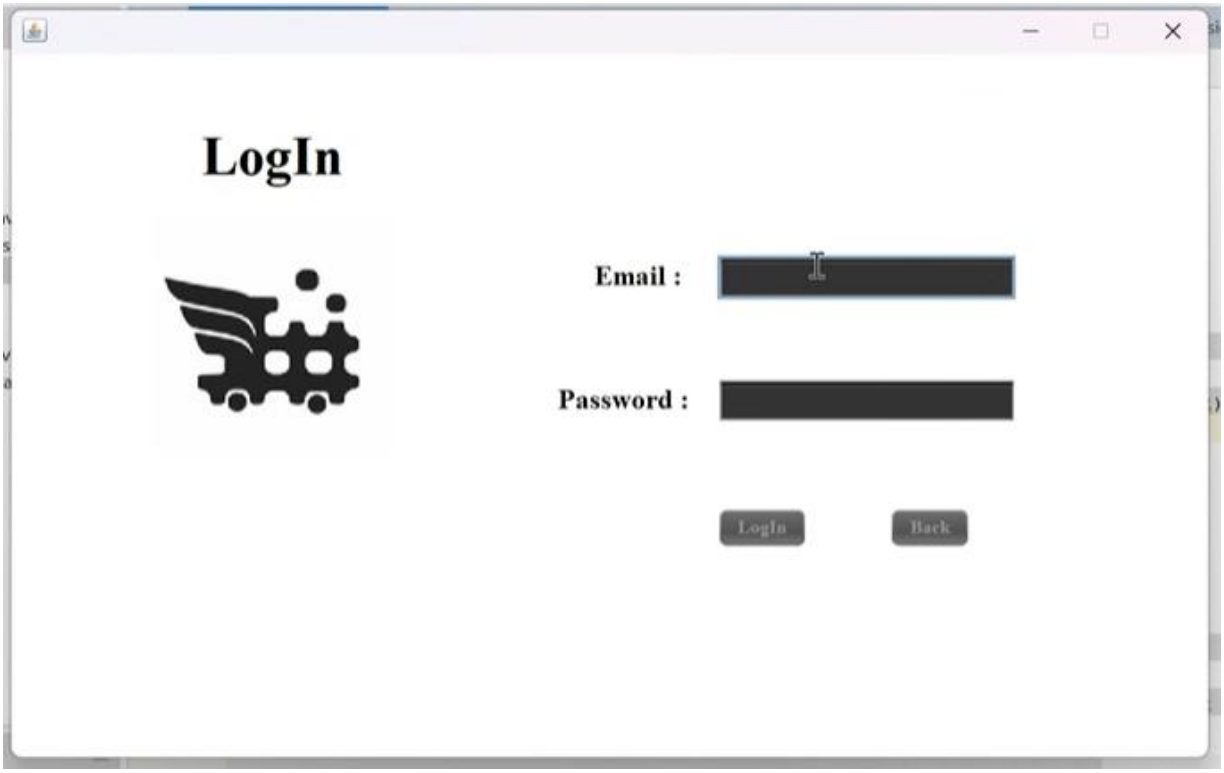
Our project interfaces



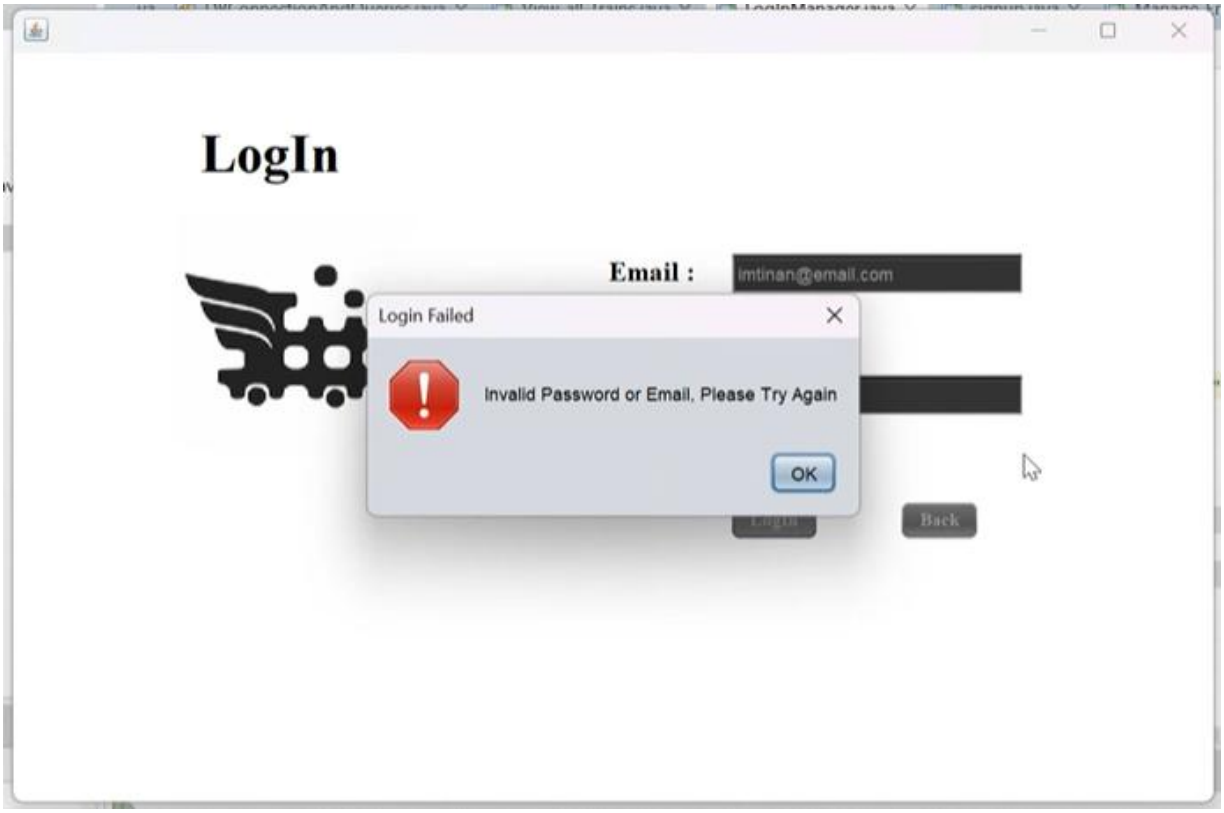
This is a home page, it will transform the user to them page.



If the user click on the Exit button the program will ask them before it closed the windows.

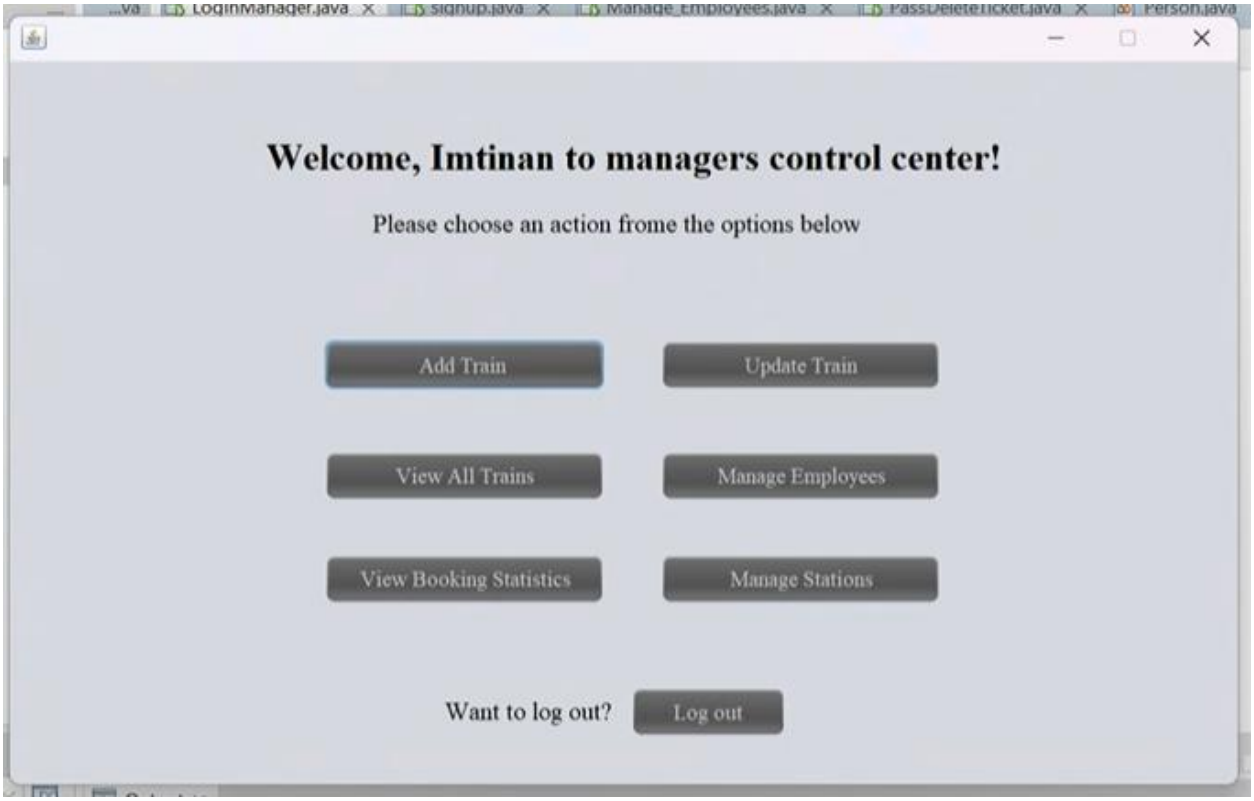


This is the manager's home page.

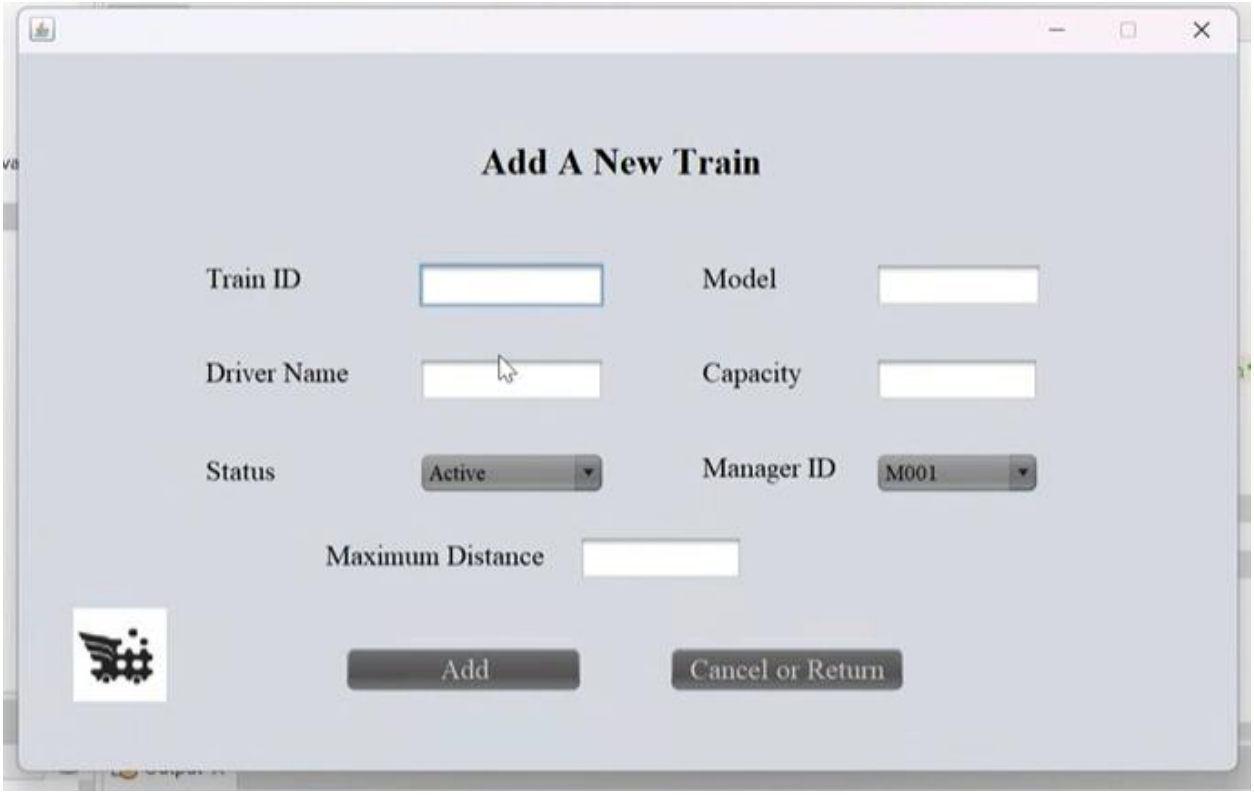


If the manager enters a wrong email, or a wrong password. The program will show a message to inform them.

After they enter a right password it will inform them that the Login successful.



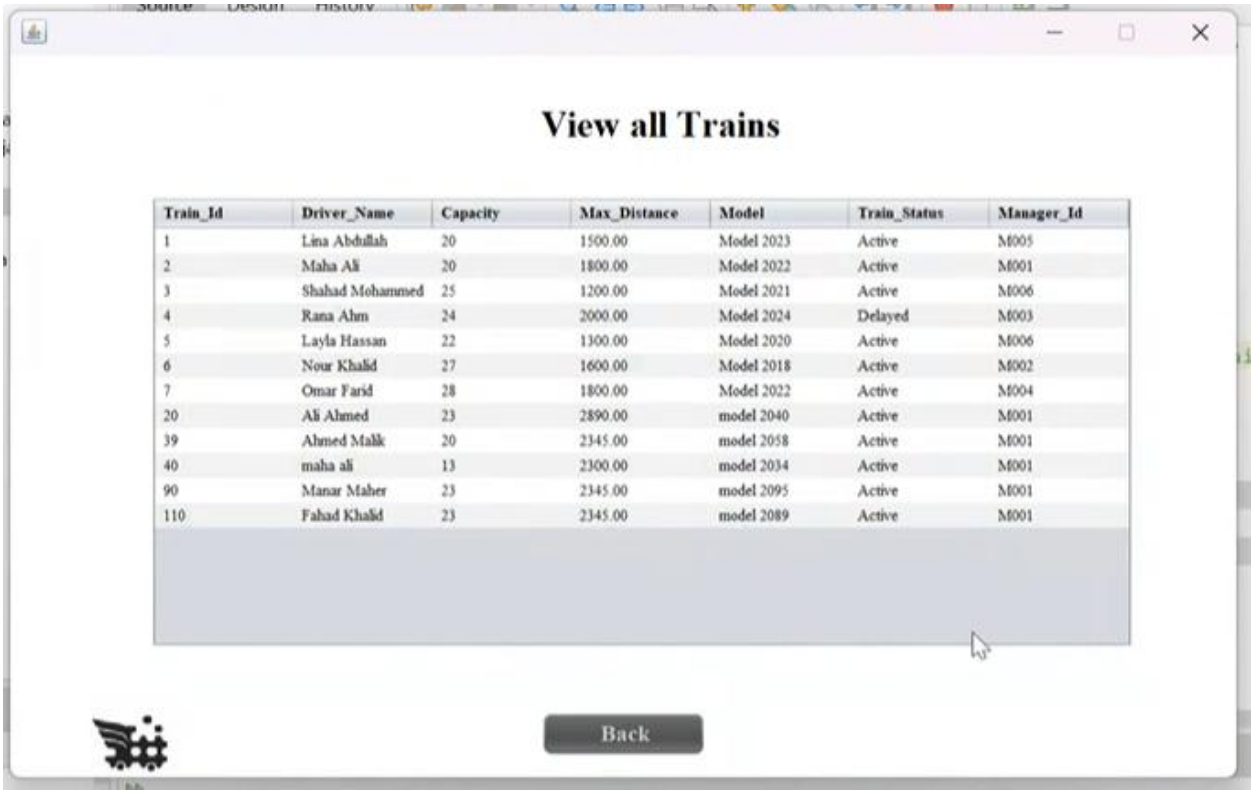
For the manager control center these are the four optional for the managers.



For Add A new train interface, the manager should fill the new train details. And choose from the JCompobox the Status and the manager Id for the train.

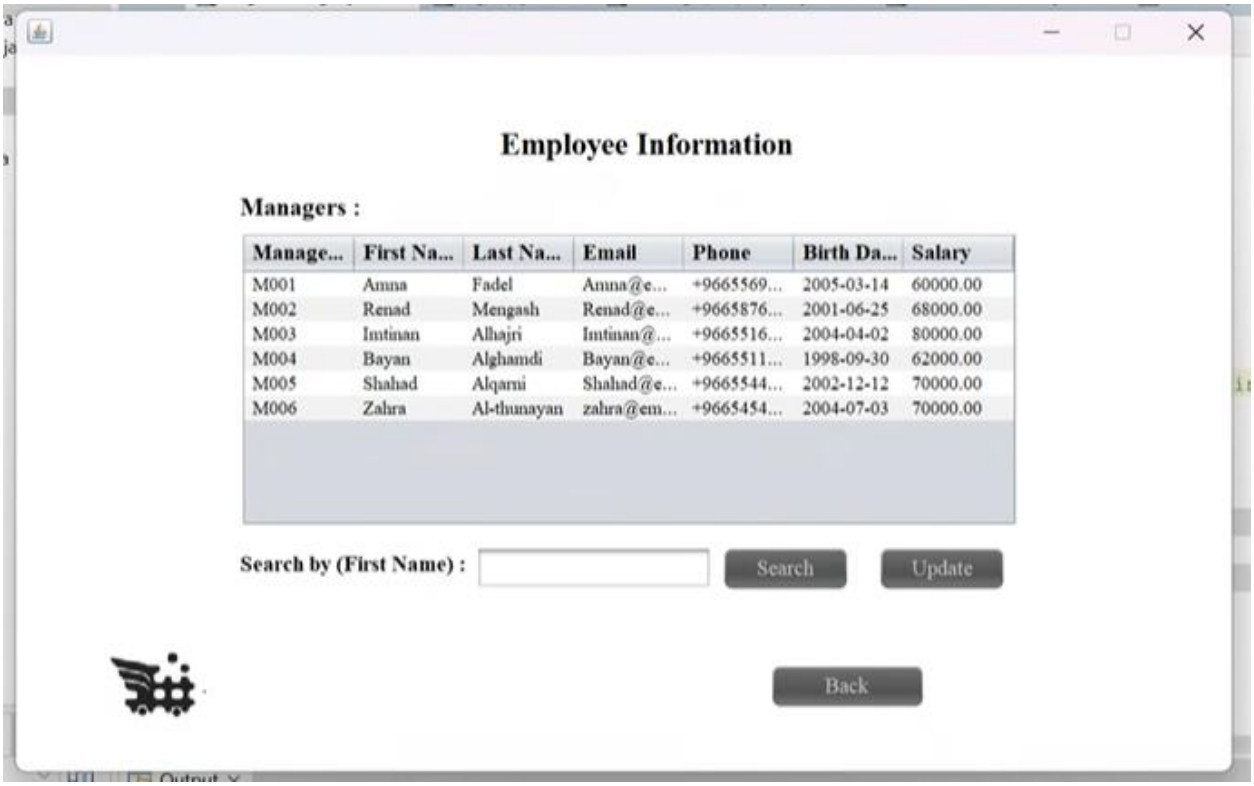
If the manager enter train Id for a train already exit, the program will inform them.

After the manager enter a successful information of the new train and click Add button the program will save the new train and inform the



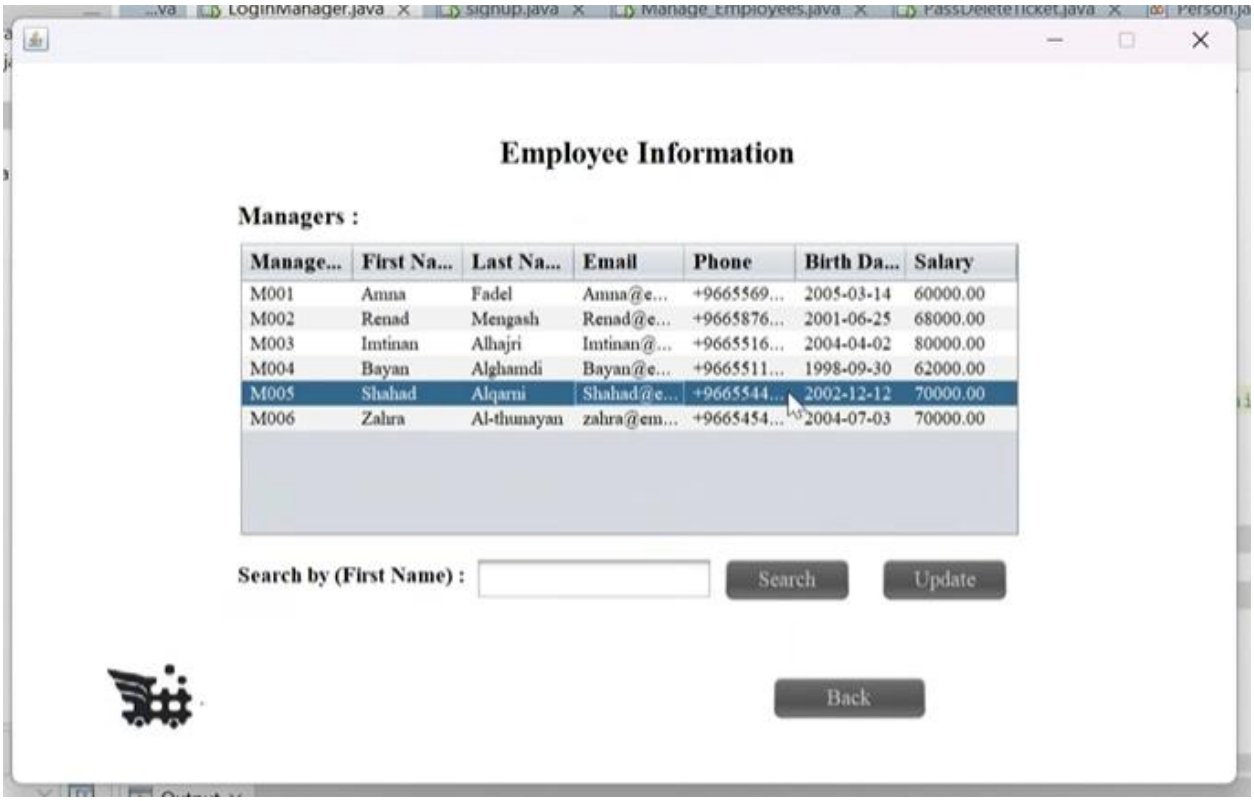
Train_Id	Driver_Name	Capacity	Max_Distance	Model	Train_Status	Manager_Id
1	Lina Abdullah	20	1500.00	Model 2023	Active	M005
2	Maha Ali	20	1800.00	Model 2022	Active	M001
3	Shahad Mohammed	25	1200.00	Model 2021	Active	M006
4	Rana Ahm	24	2000.00	Model 2024	Delayed	M003
5	Layla Hassan	22	1300.00	Model 2020	Active	M006
6	Nour Khalid	27	1600.00	Model 2018	Active	M002
7	Omar Farid	28	1800.00	Model 2022	Active	M004
20	Ali Ahmed	23	2890.00	model 2040	Active	M001
39	Ahmed Malik	20	2345.00	model 2058	Active	M001
40	maha ali	13	2300.00	model 2034	Active	M001
90	Manar Maher	23	2345.00	model 2095	Active	M001
110	Fahad Khalid	23	2345.00	model 2089	Active	M001

For the view all train interface, it will shown a list of all the trains details.

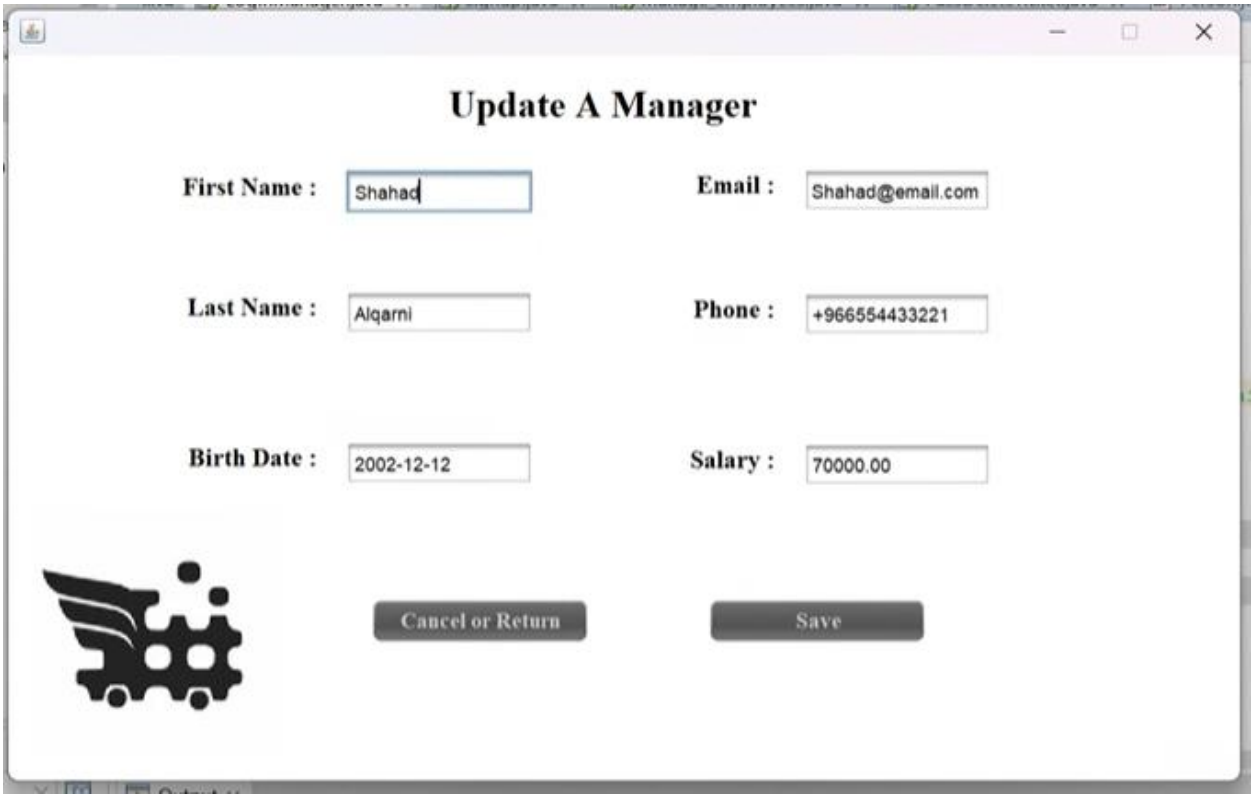


For the Employee information interface, it will shown a managers details as a list.

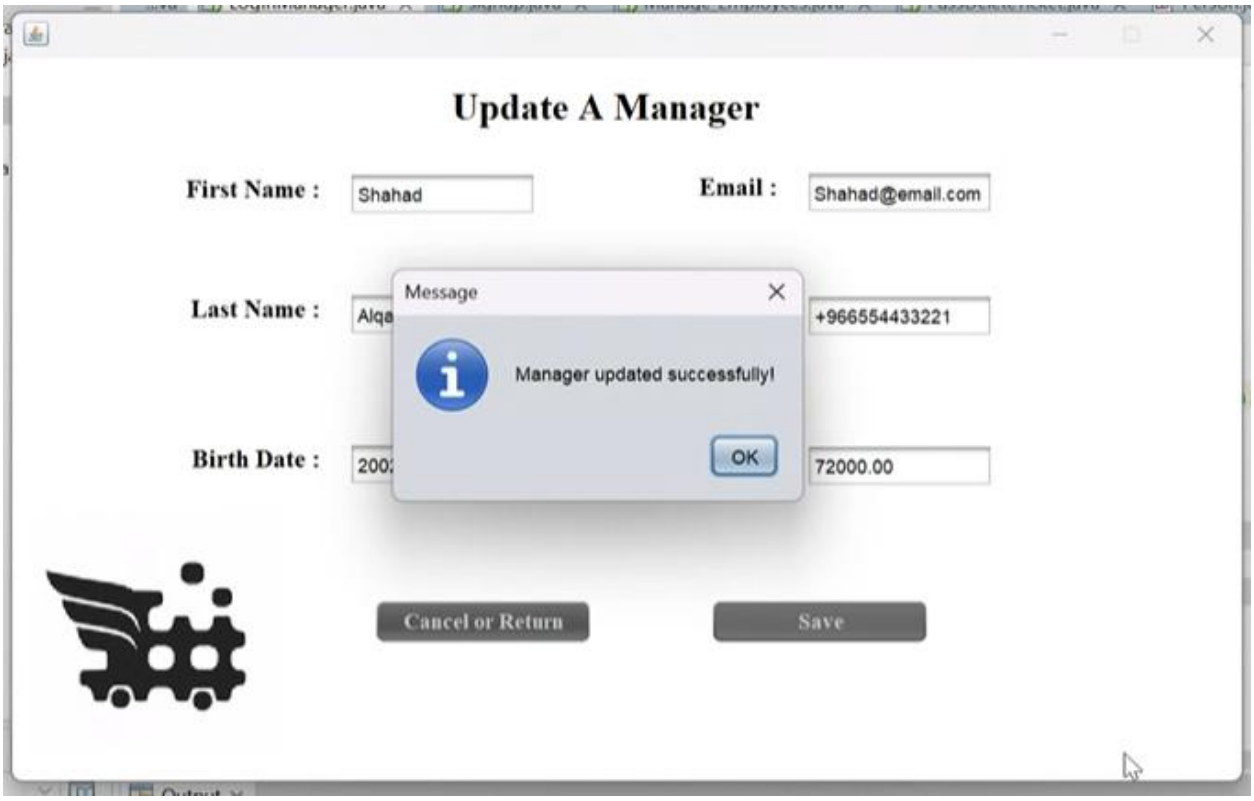
The manager can search for someone by their first name.



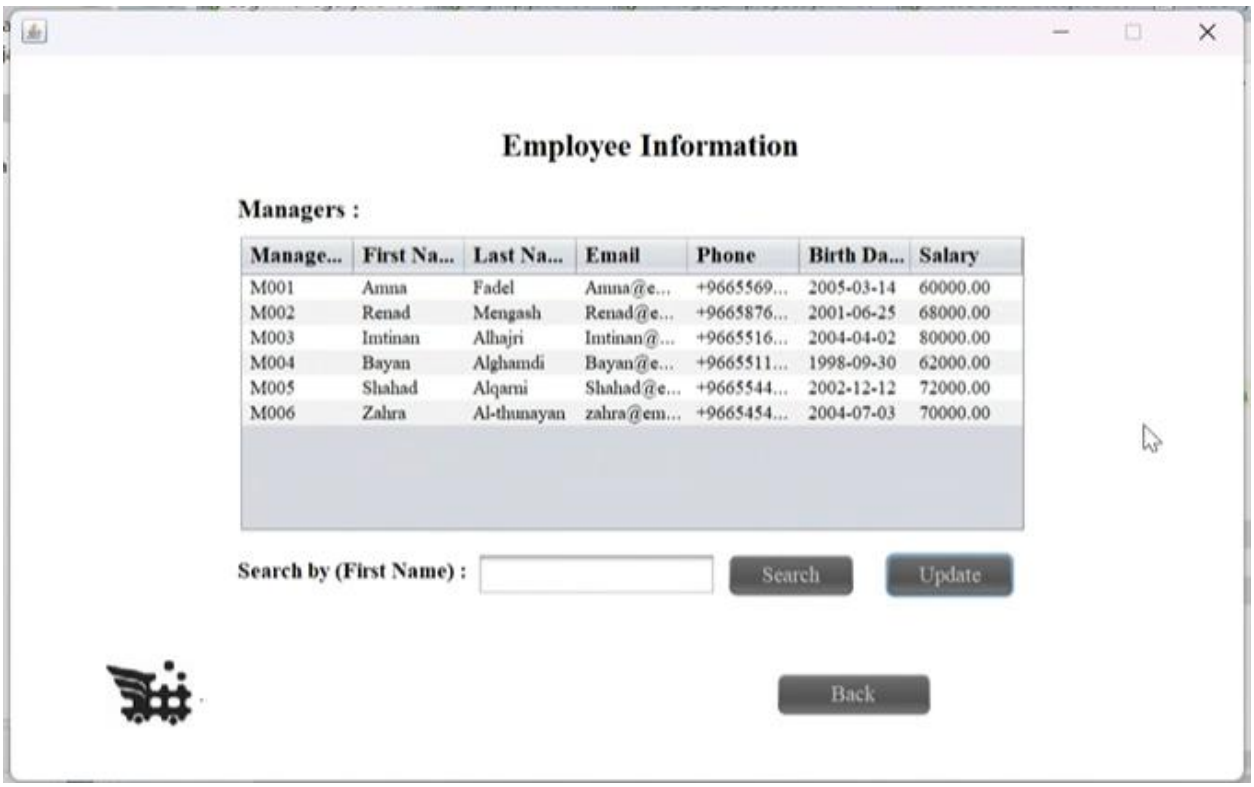
The manager can Also select a specific managers to update their information.



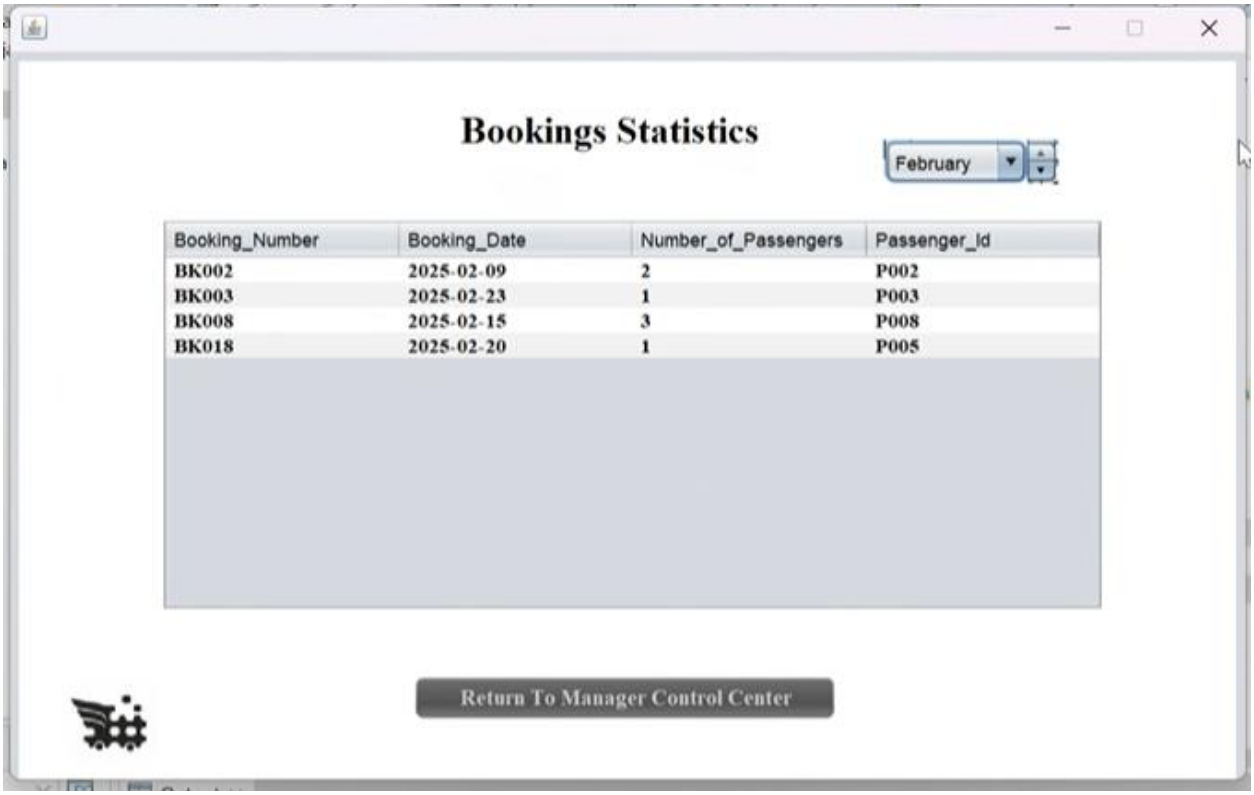
This is the Update a manager interface that filled automatically by the program.



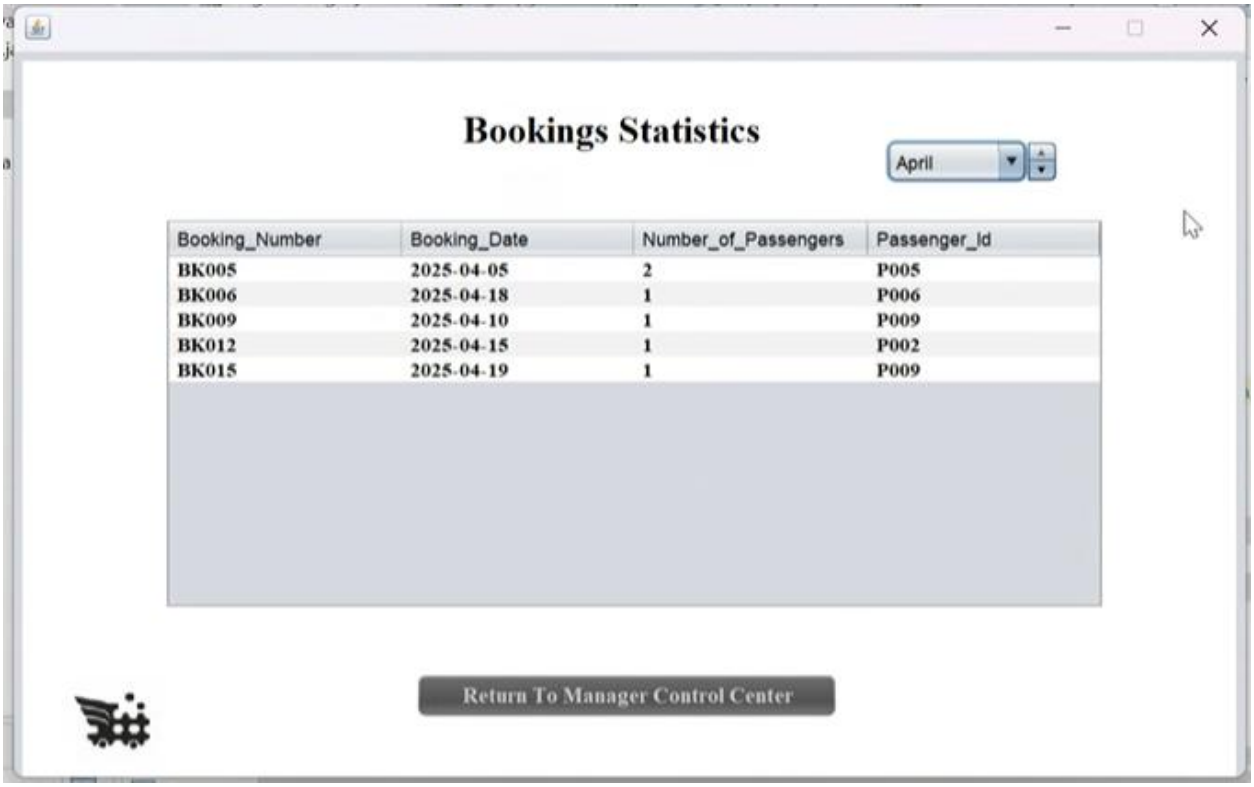
After the manager update an any information of the other manager, and click Save, the program will inform them that the update is succssfully.



Employee Information after updating.



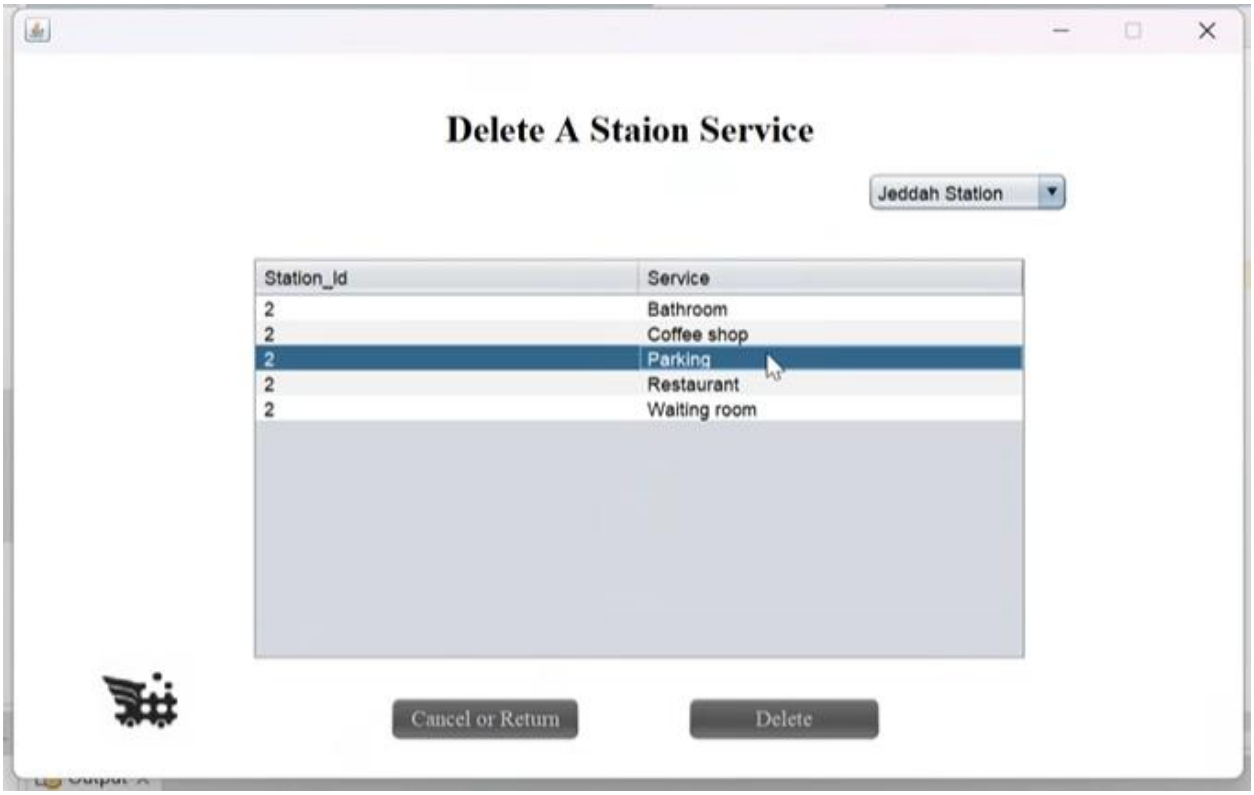
A view booking statistics interface will shown the booking details for the selected month.



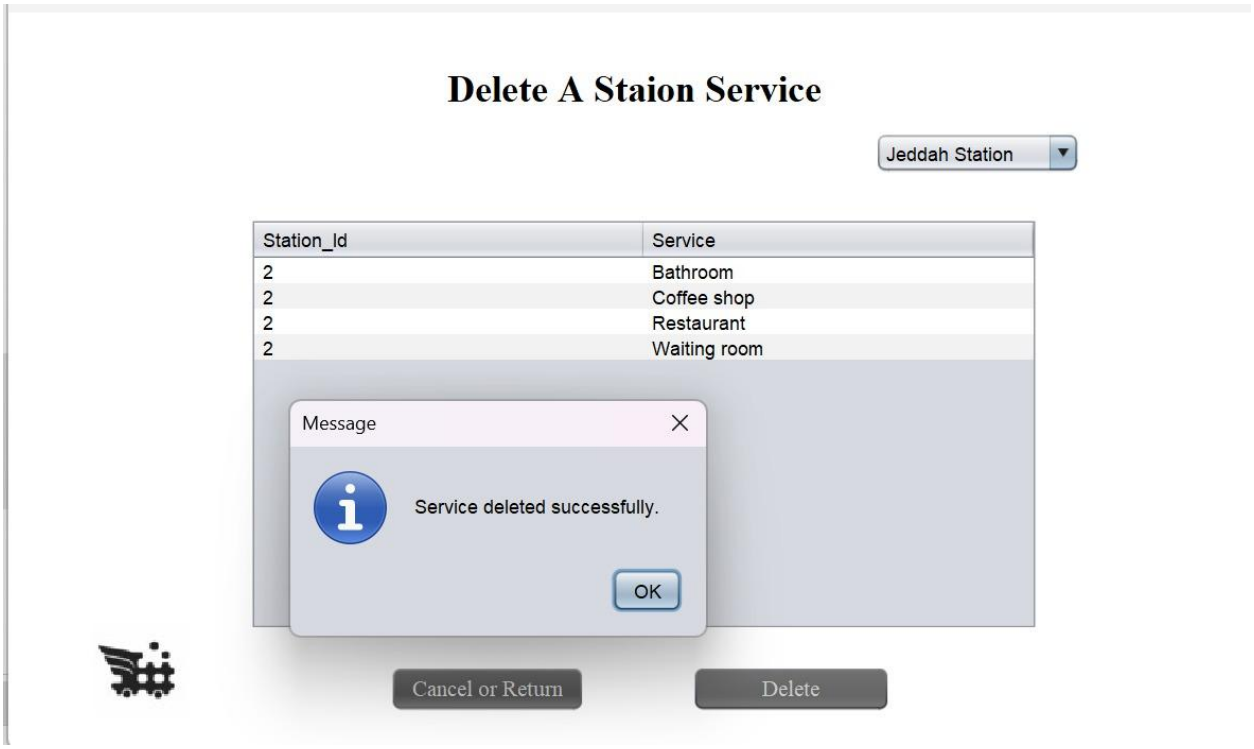
Another month Selected.

The Delete A Station Service let the manager choose one Station from the JComboBox.

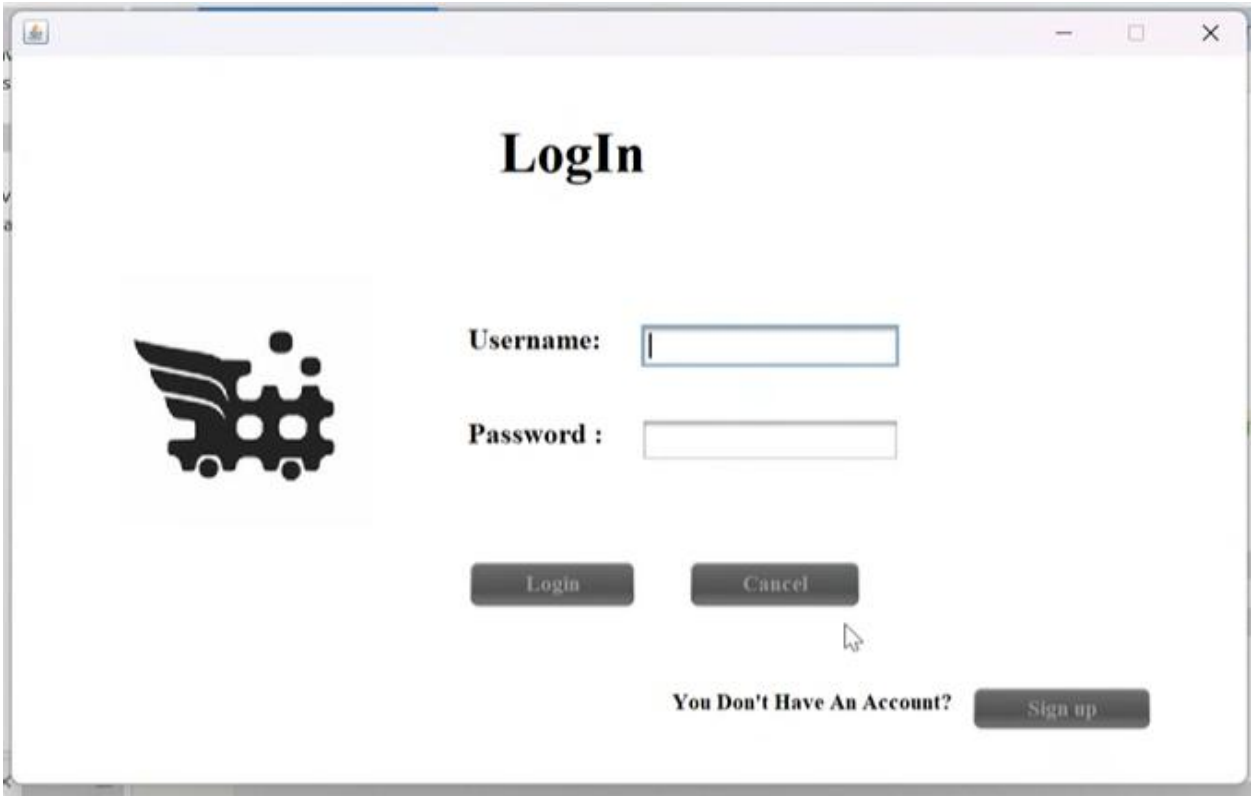
All the Services of the Selected Station.



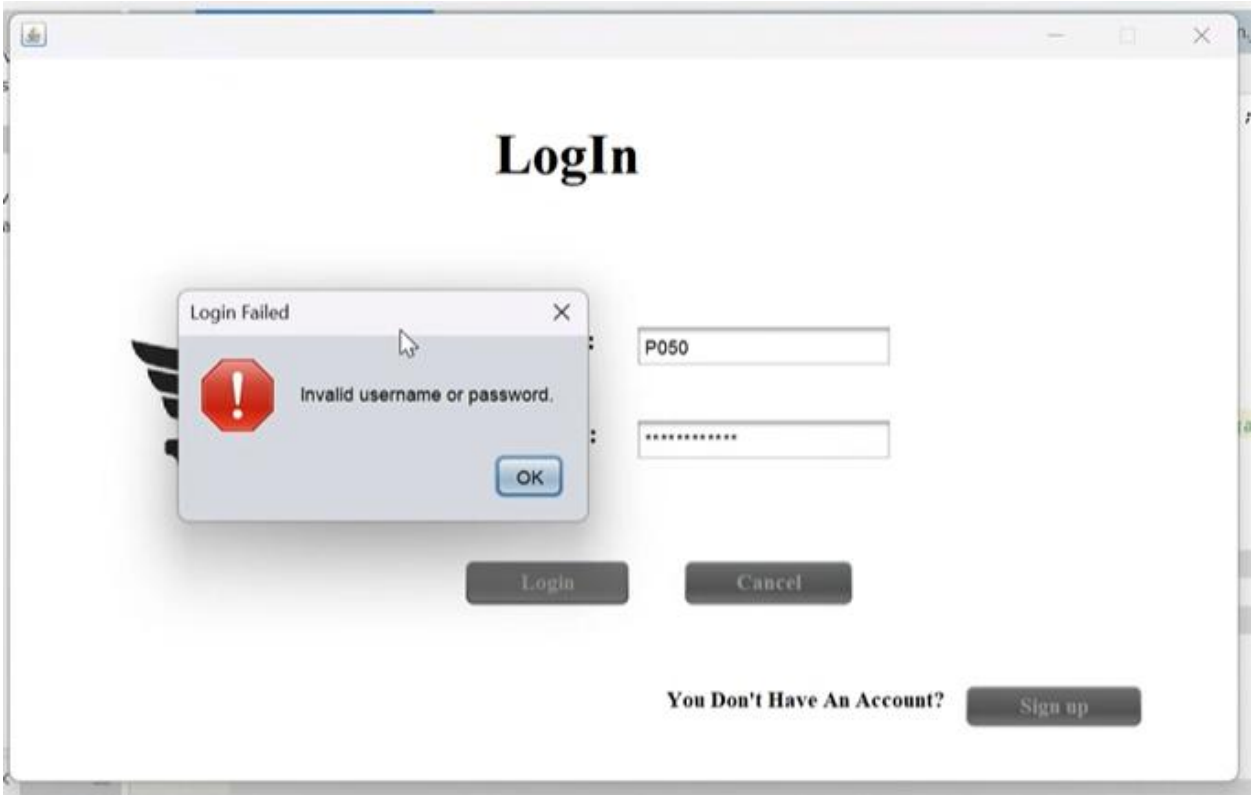
Select one Services To delete it.



Service deleted successfully.



This is the passenger home page.



The program will inform the passenger when they enter a username or password that didn't found on the database.

Sign Up interface will required the user to fill their personal information. And inform them the correct formal of the password by using tool Tip.

The passoword will return red stars Until the user enter a successful password.

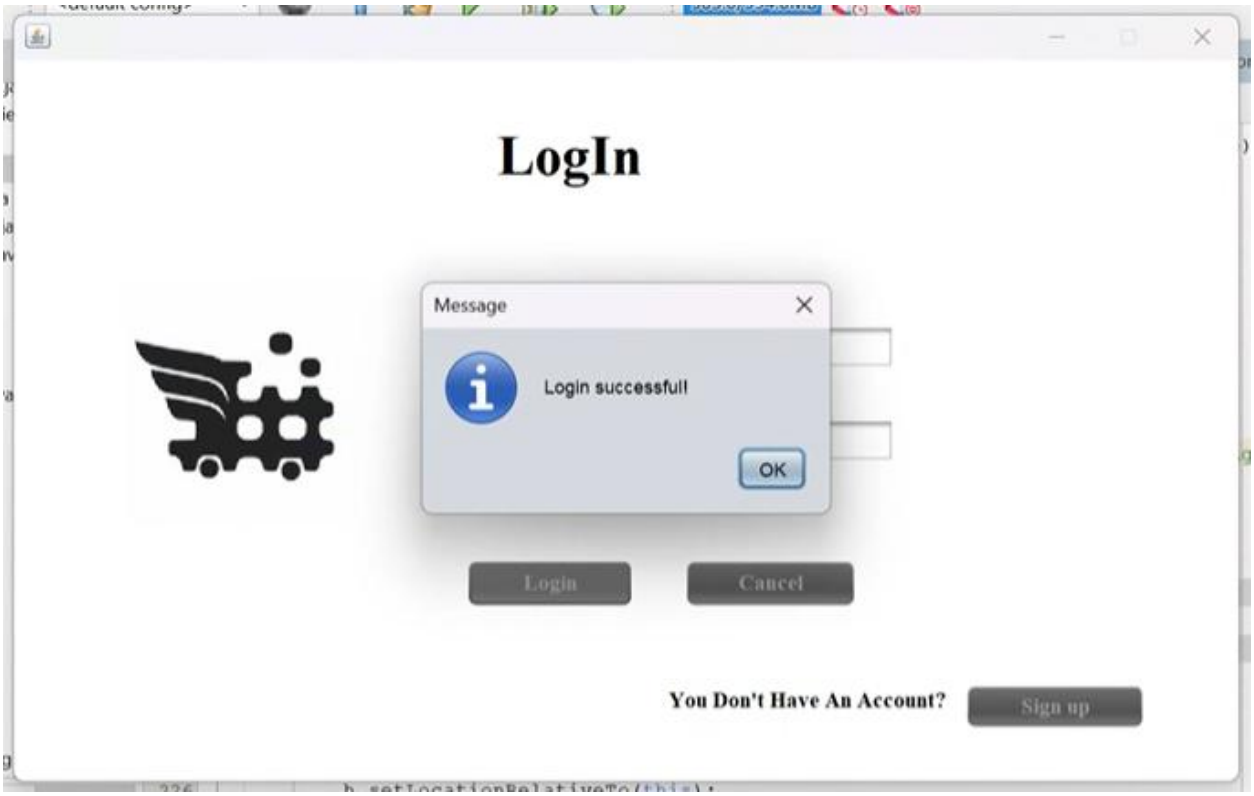
It will inform them if they didn't enter a valid password.

If the user was under than 18 years old, they cannot Sign up.

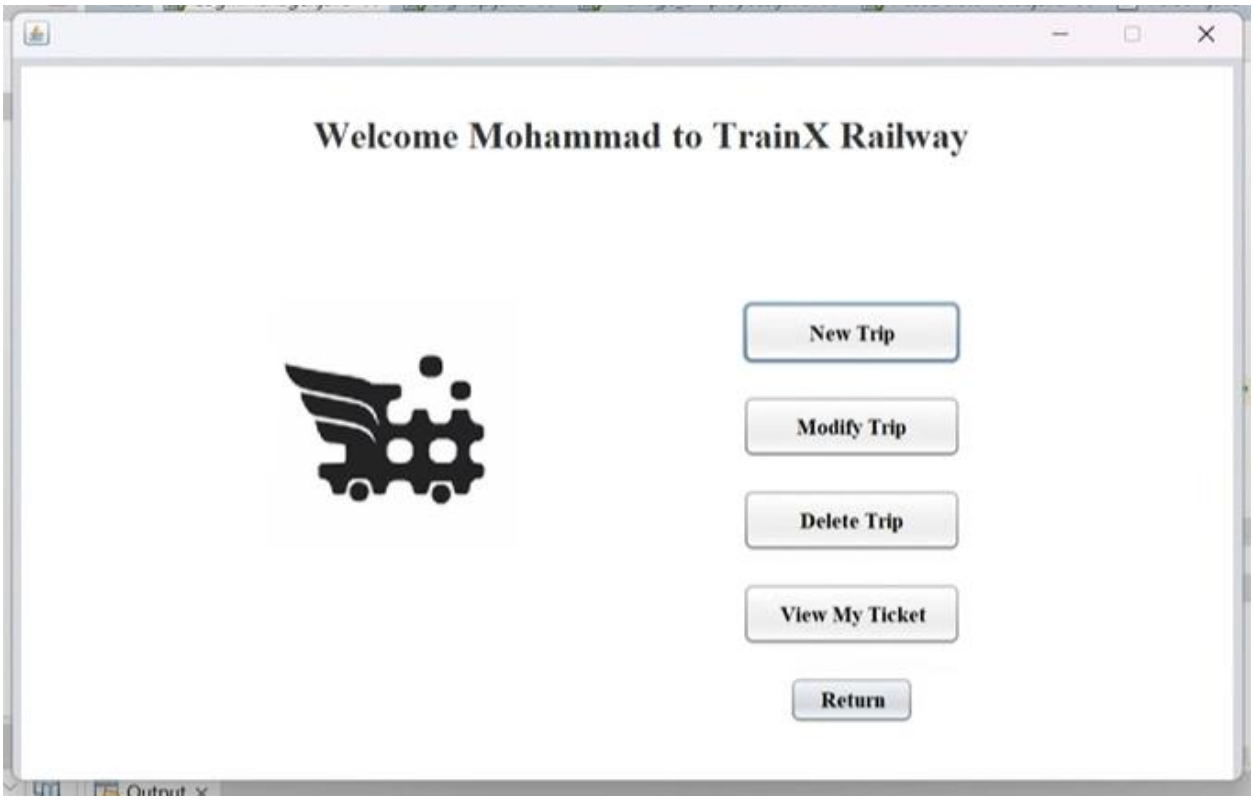
The birth date should be in the past.

After enter a valid information and click on the Sign Up button, the Sign up Successful.

If the user fill the information of the user is already found in the database, the program will ask them if they want to Sign in.

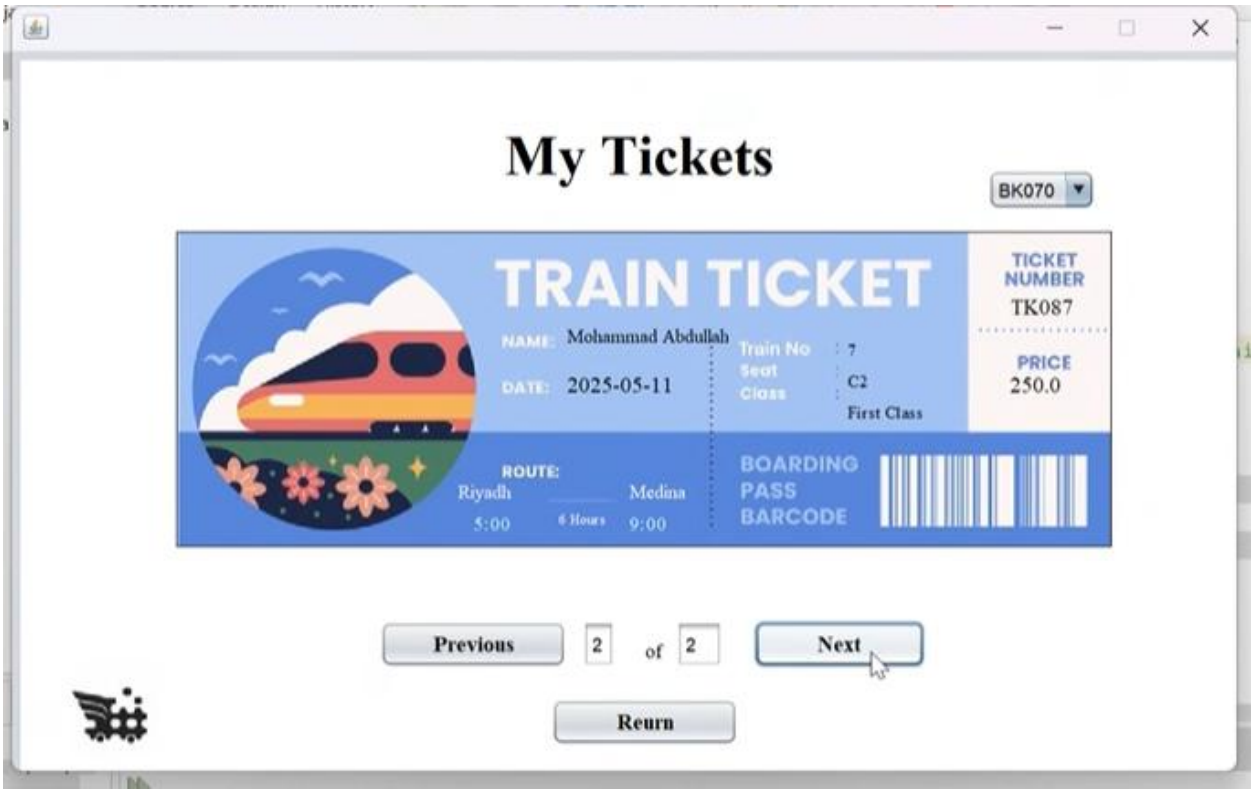


If the passenger enter a valid username and password it will inform them the Login successful.

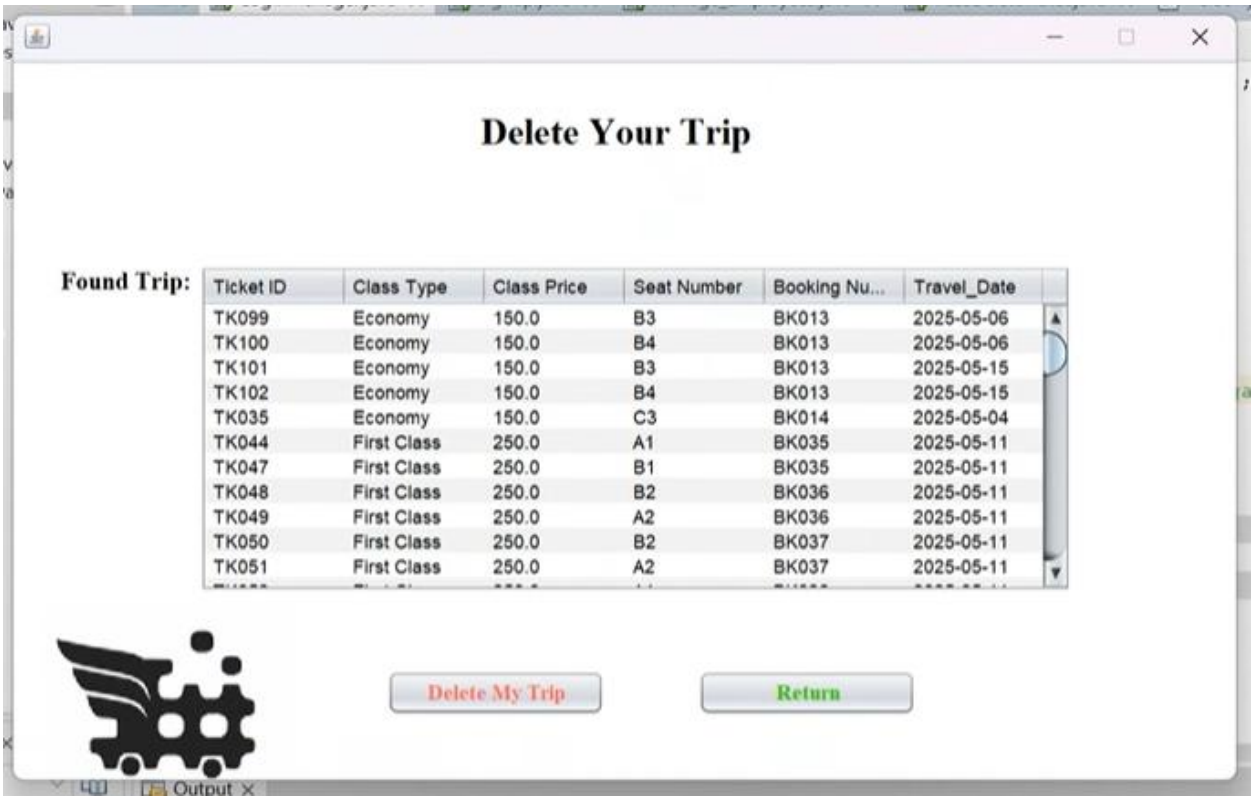


In the passenger home page, it will show their name in the title and let them choose one from four operations.

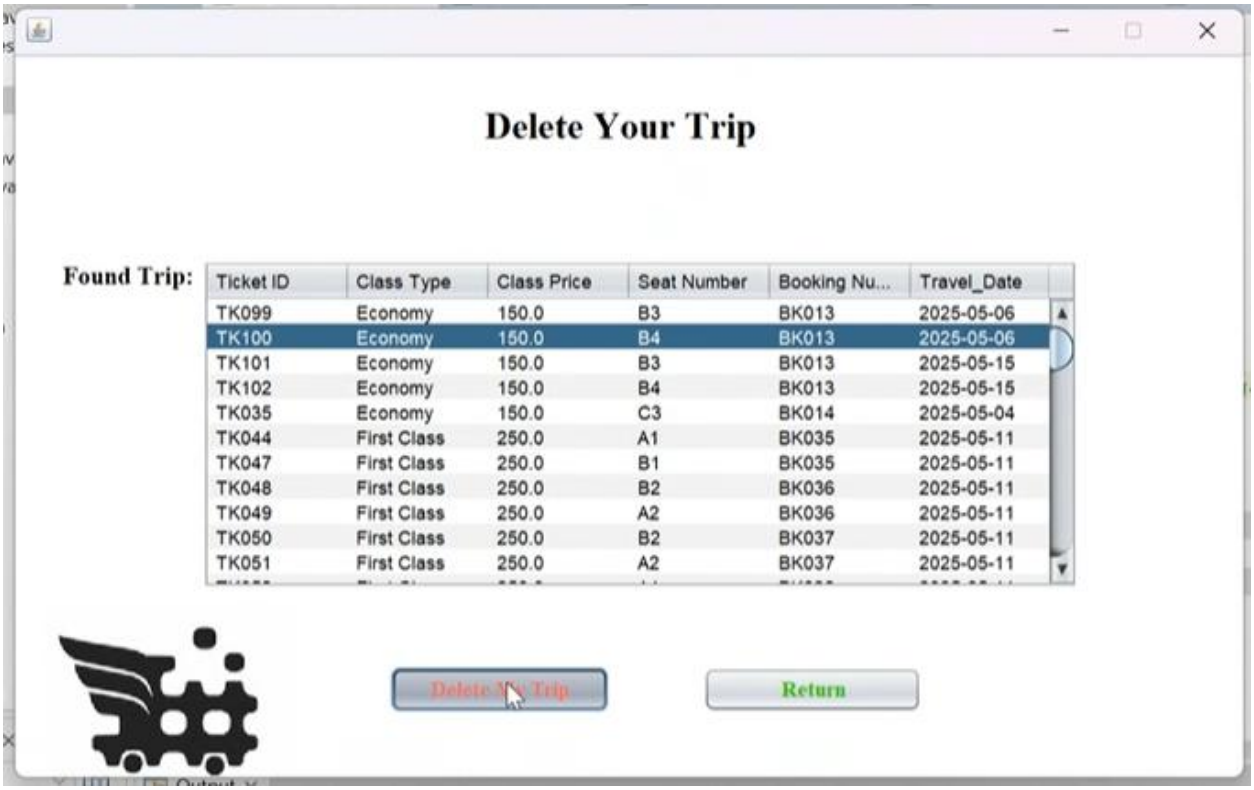
The View my Ticket interface will print the trip detail based on the booking number.



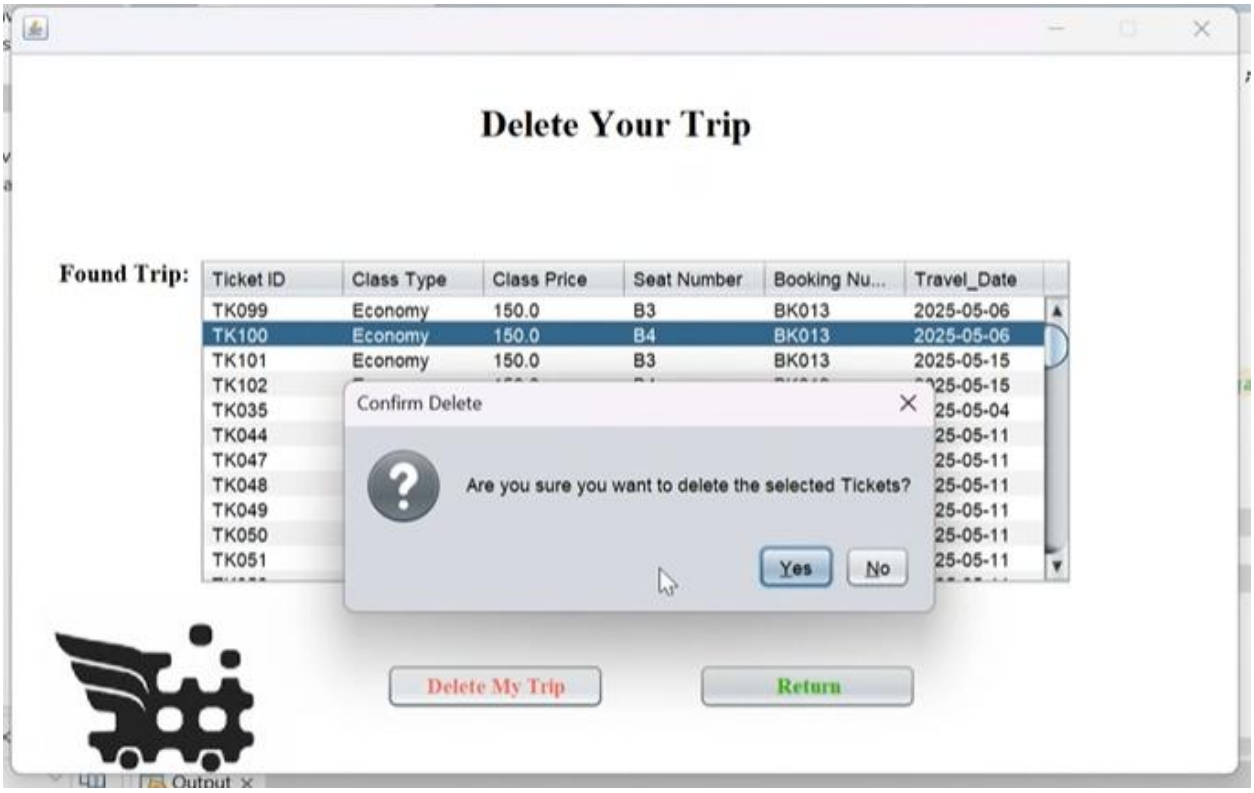
Another ticket for the same trip.



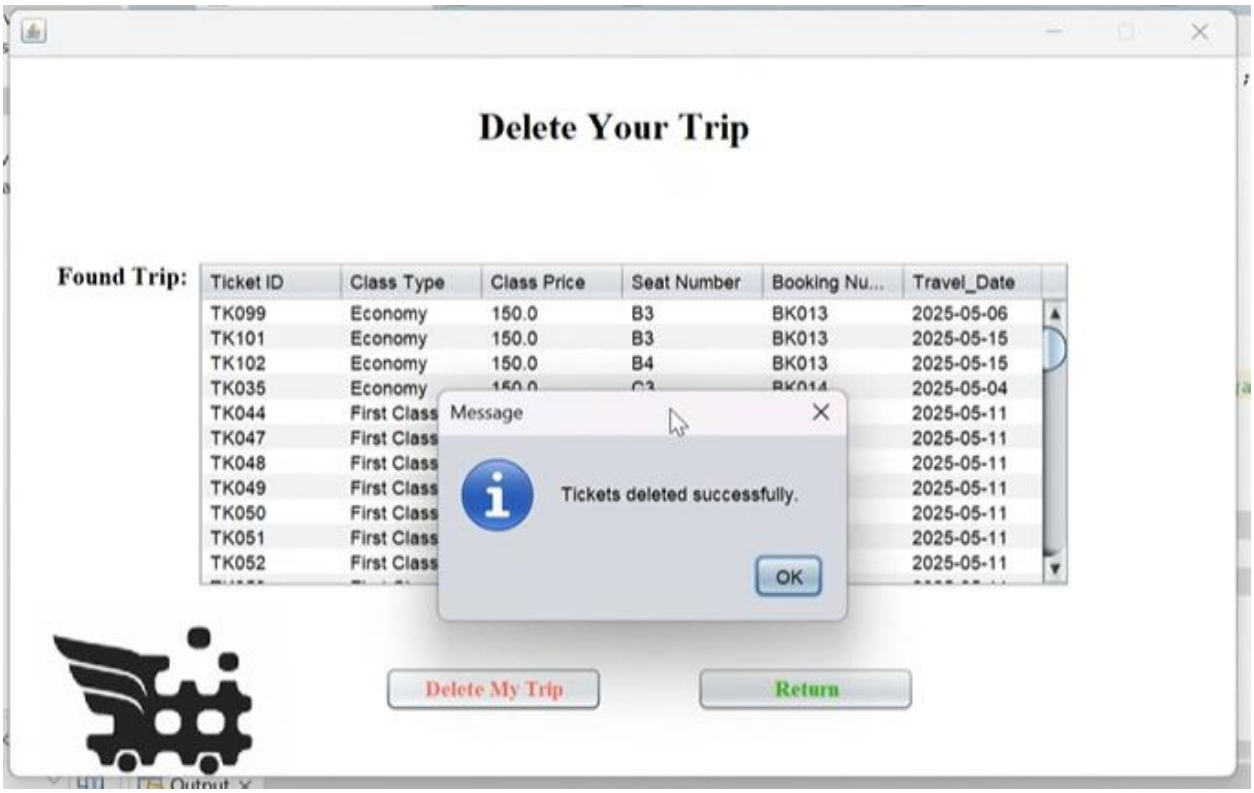
For the Delete Your trip interface it will show them a list of their ticket.



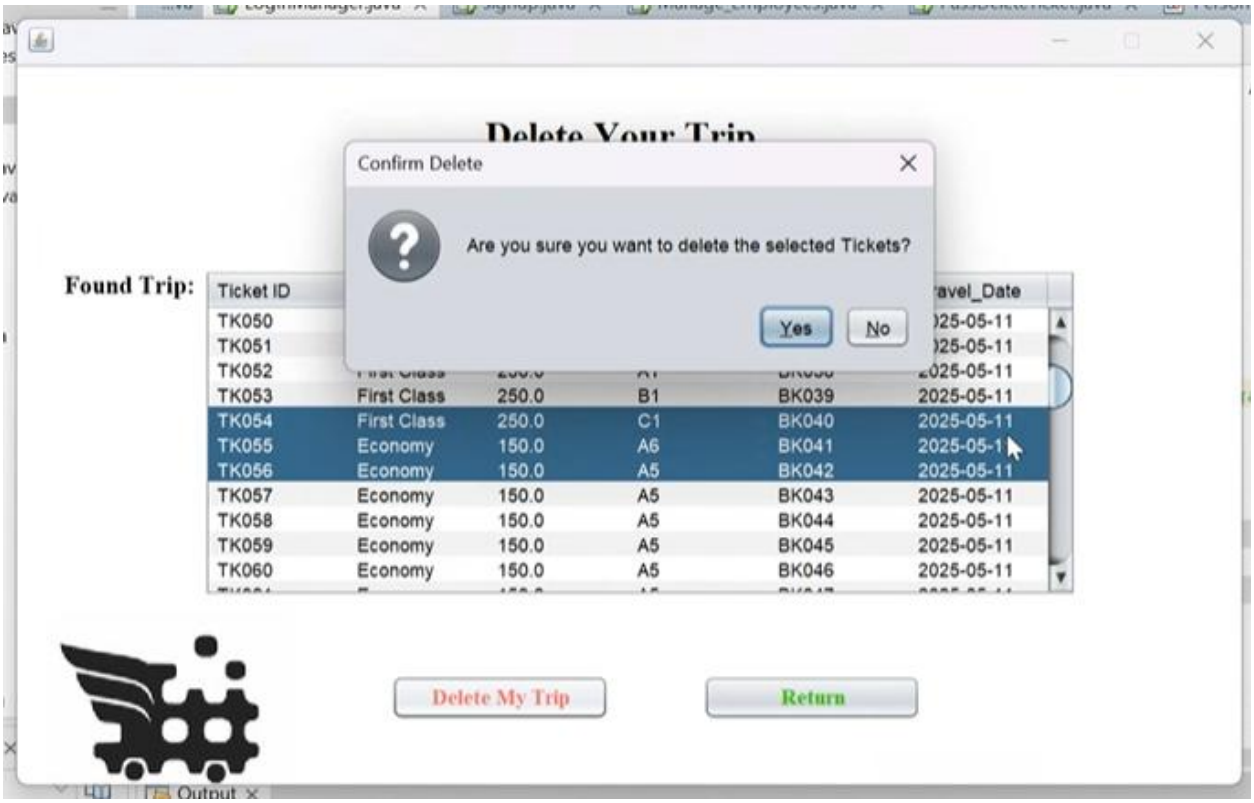
They can select one Ticket of them to delete it.



The program will ask them if they are sure before deleting the Ticket.



If they click ok, the Ticket deleted successfully.



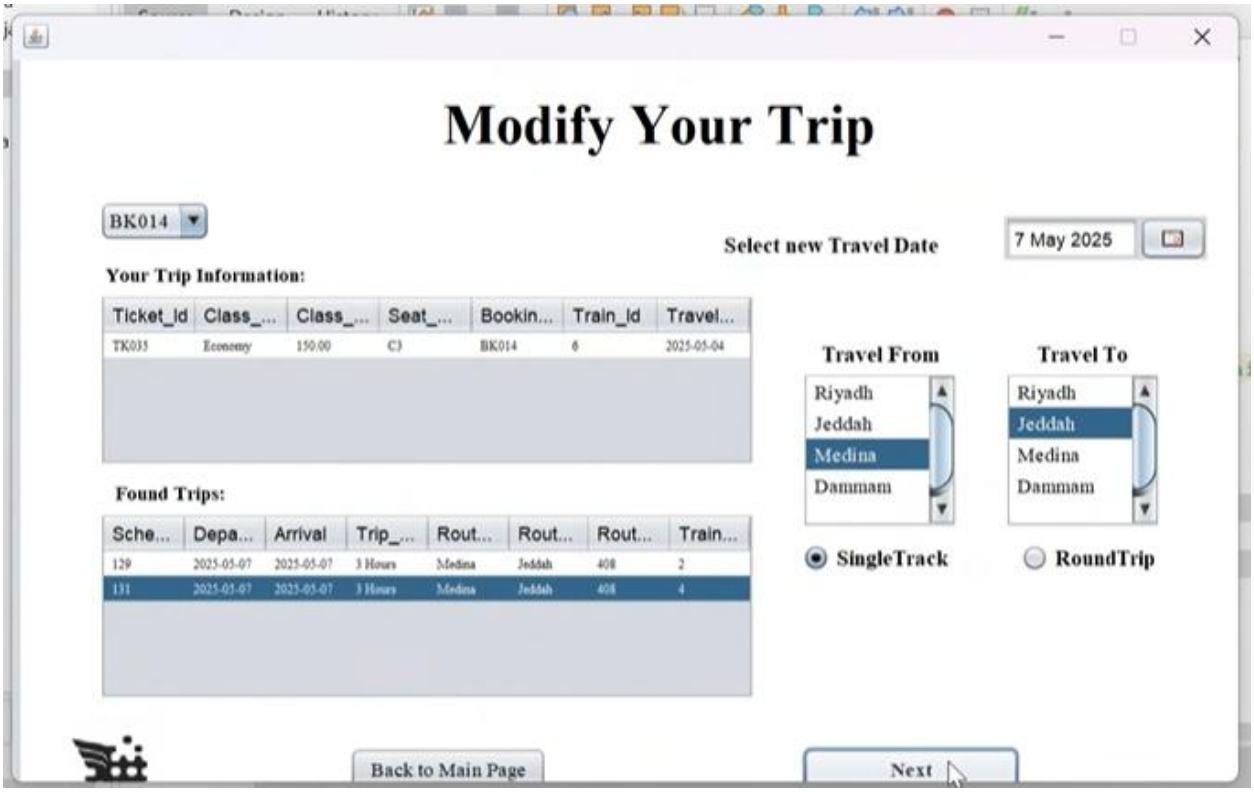
They can Also delete multiple Ticket at once.

Modify Your Trip interface will show the trip details based on the booking number. And let the user fill the new trip details.

If there is no found trips with that details, the program will show a label say Sorry.

If the user Select a some city to travel to, The program will show an information message.

When the user enter a valid trip details for a trips already exit in the database, the found trips will shown in the list below.

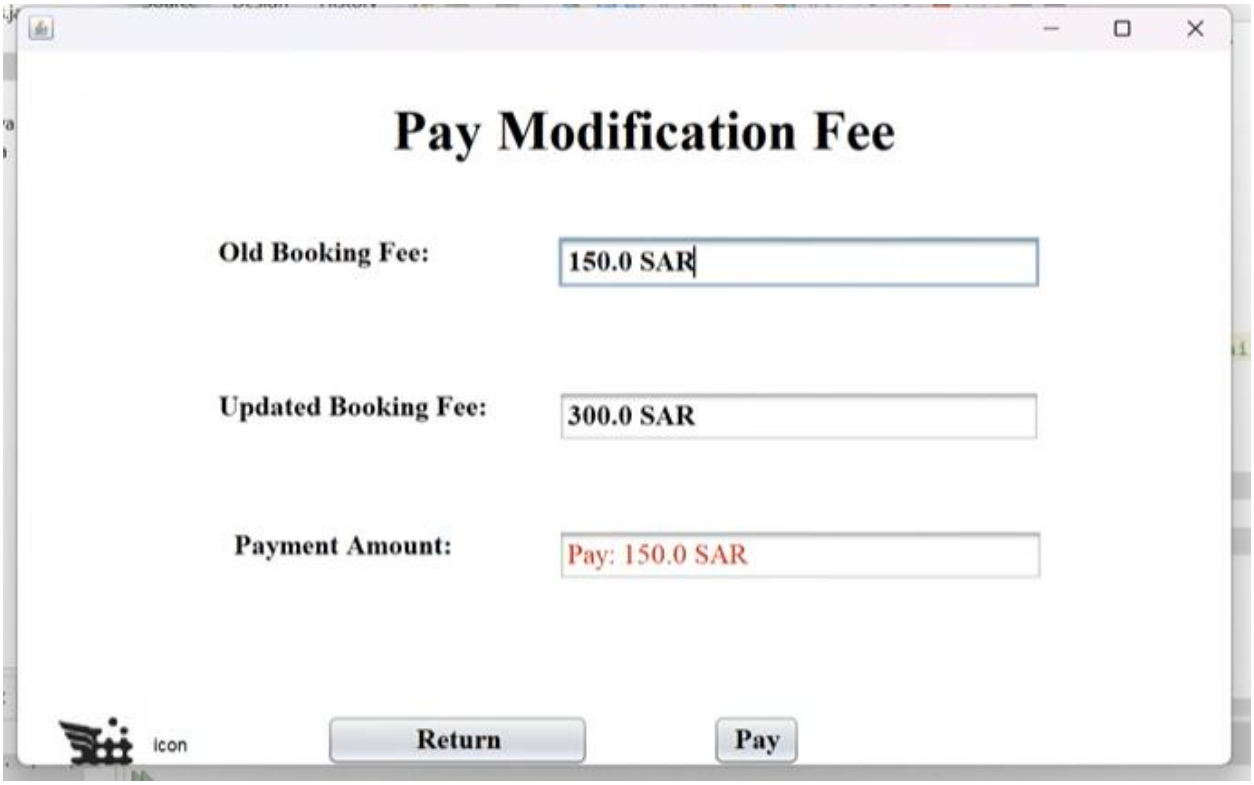


They can Select one of them.

Selecter Your new Trip Details interface.

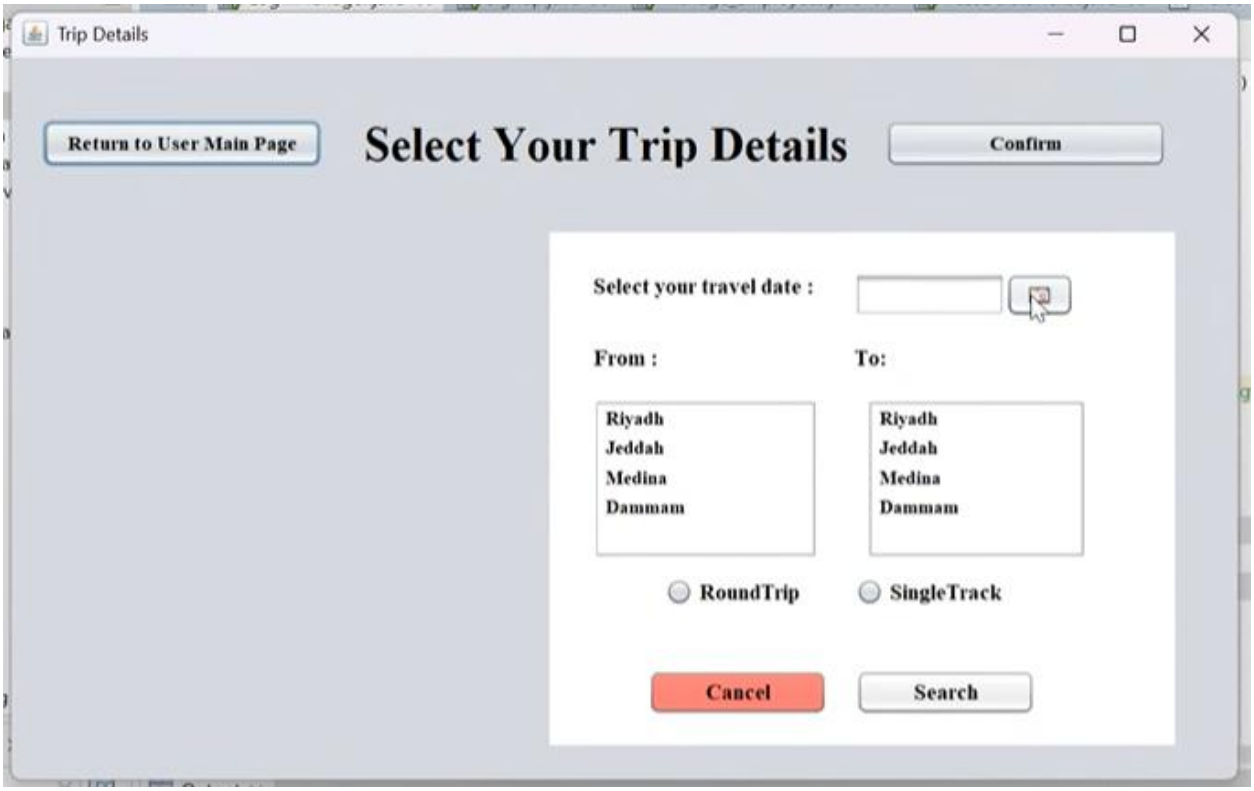
hen Choose they Seat from the panel Seats and it will print the Selected Seats with the Total Fee automatically.

When they click on the Confirm button, the program will ask them to Proceed for payment.

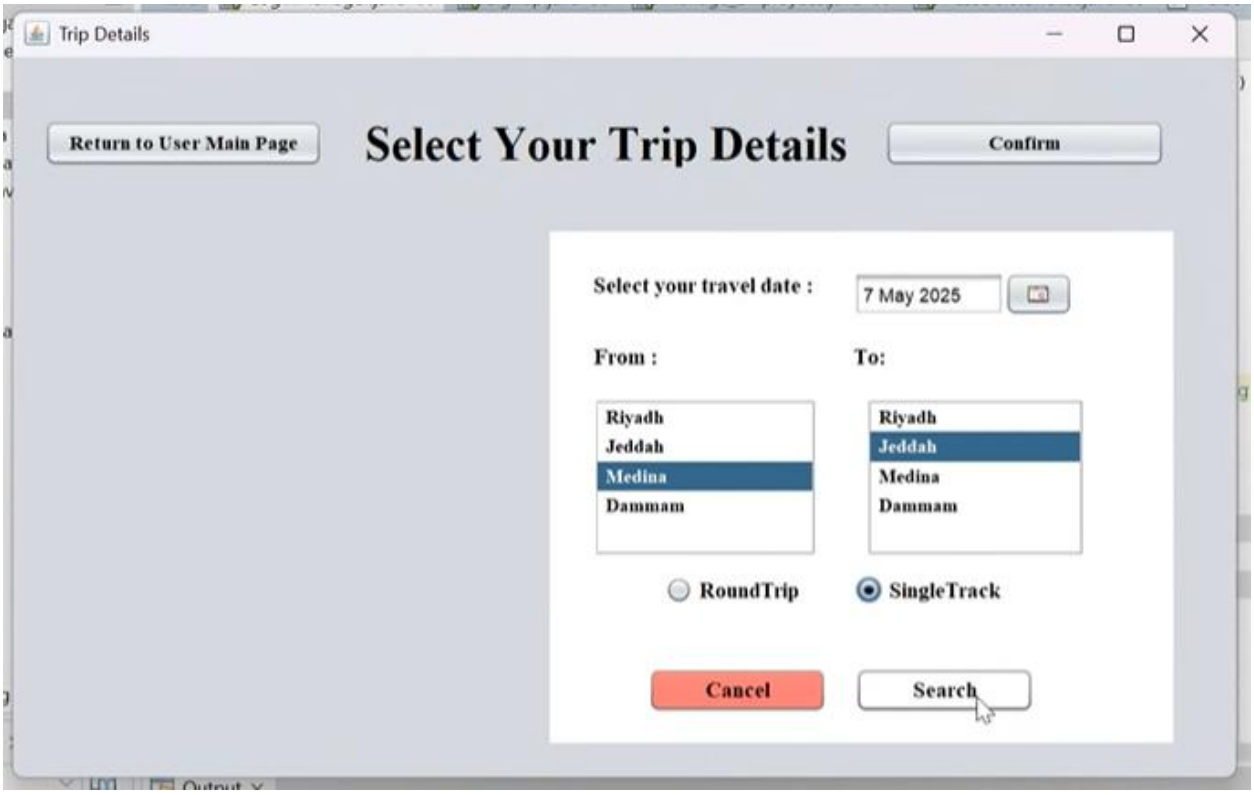


Payment Details After modify you trip interface.

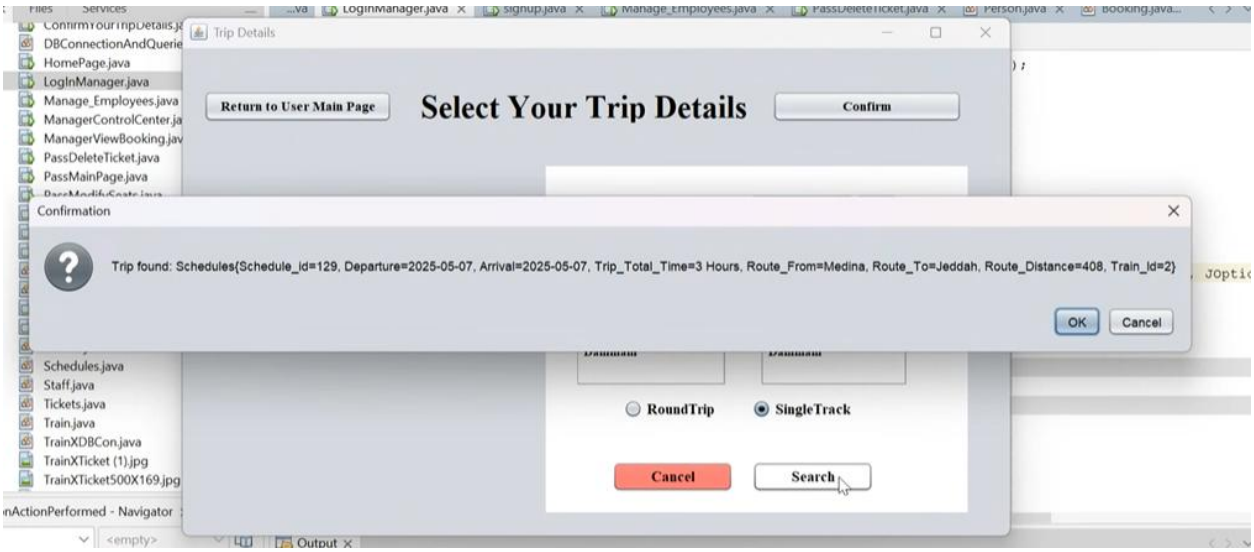
When you modify your trip the program will till you to come to the nearest station to completed the payment.



For the booking New Trip interface, it will show the right panel first with the trip details.



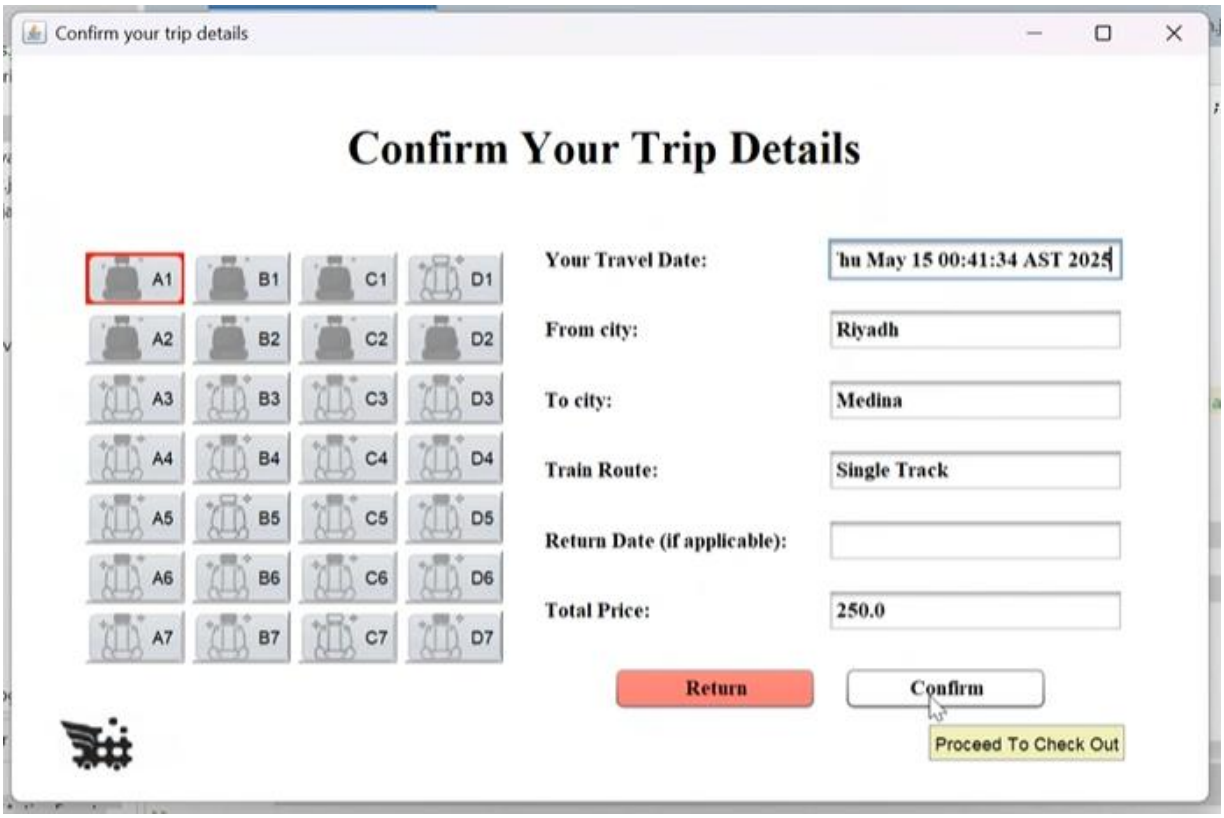
The user fill the trip details then click Search (if the user choose a Round Trip, JchooseDate will shown belos).



If there is a trip with that details, the program will inform the user all about that trip information.

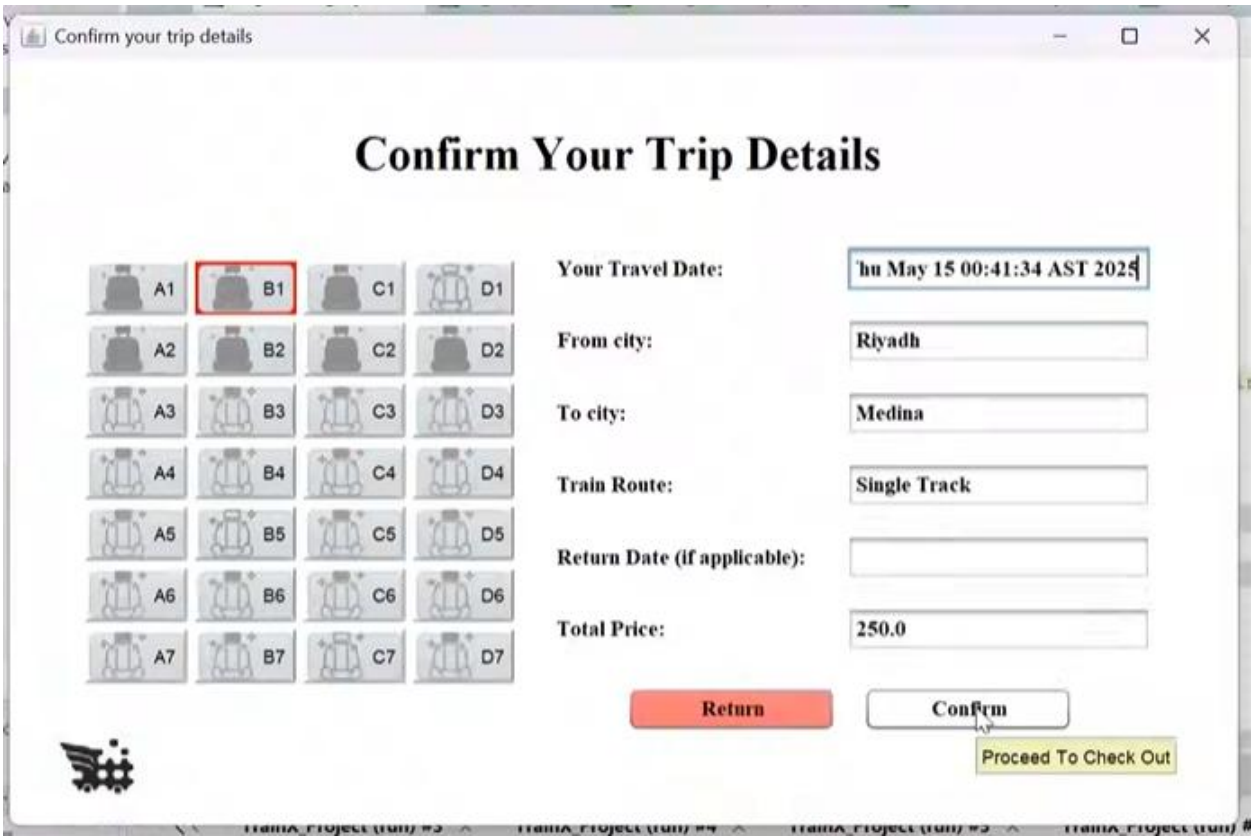
Then the trip details will be uneditable and the Seat Selection panel will show.

They can Choose their Seat.

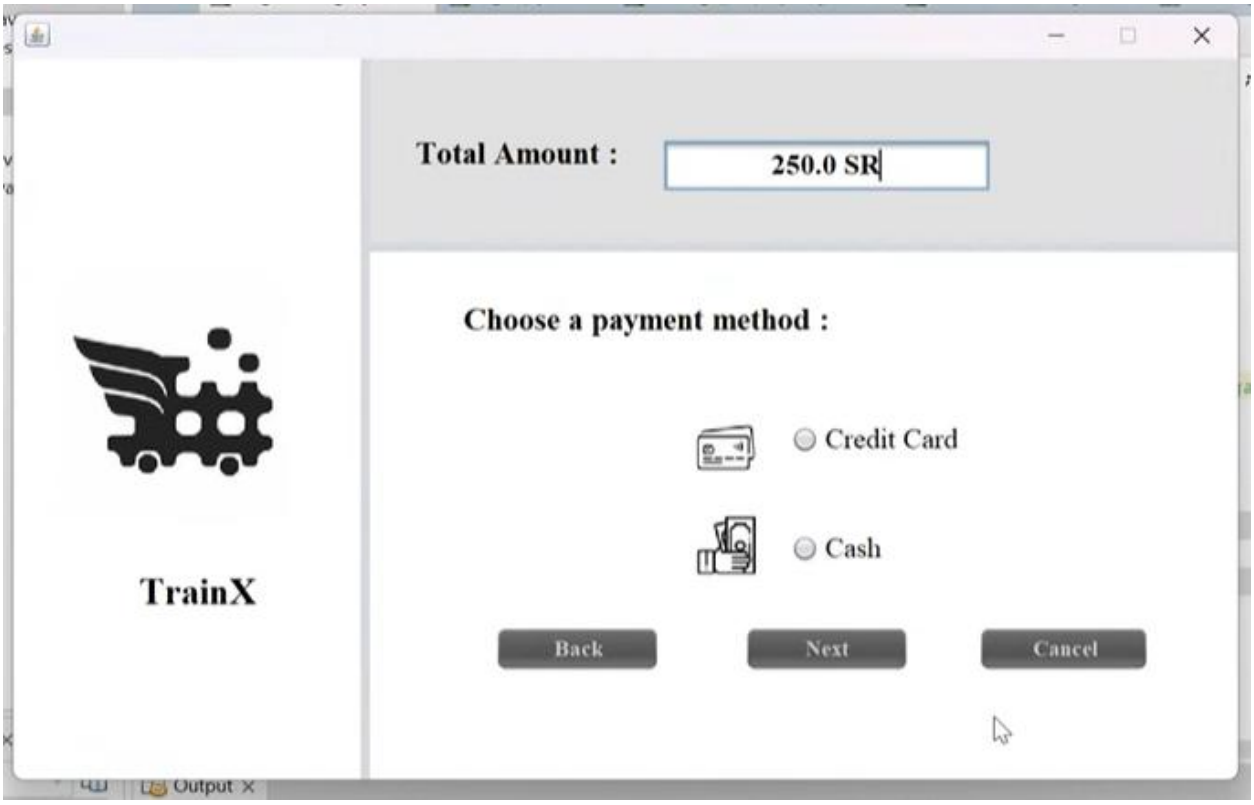


After Click Confirm button the Confirm Your Trip Details interface will show the passenger all their trip details with the Total price.

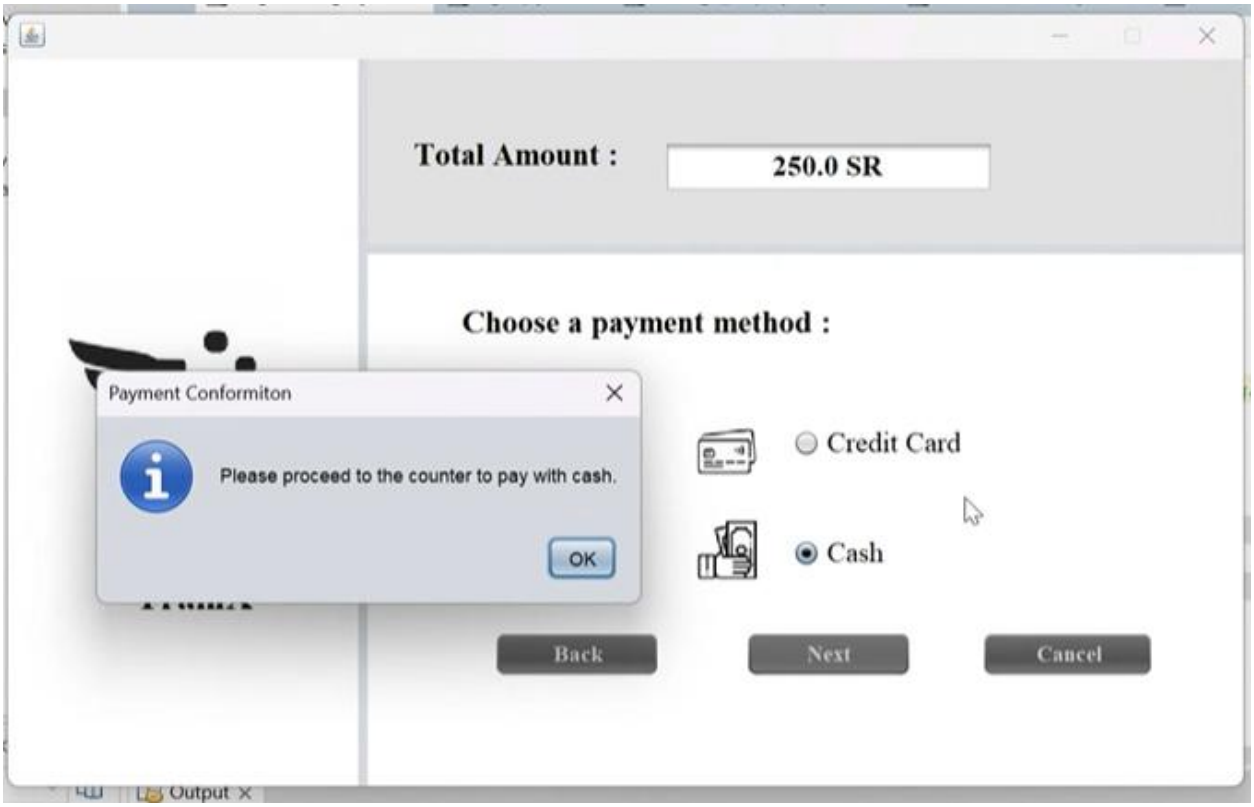
If the user Click Return in the Confirm Your Trip Details interface, they can change their seat.



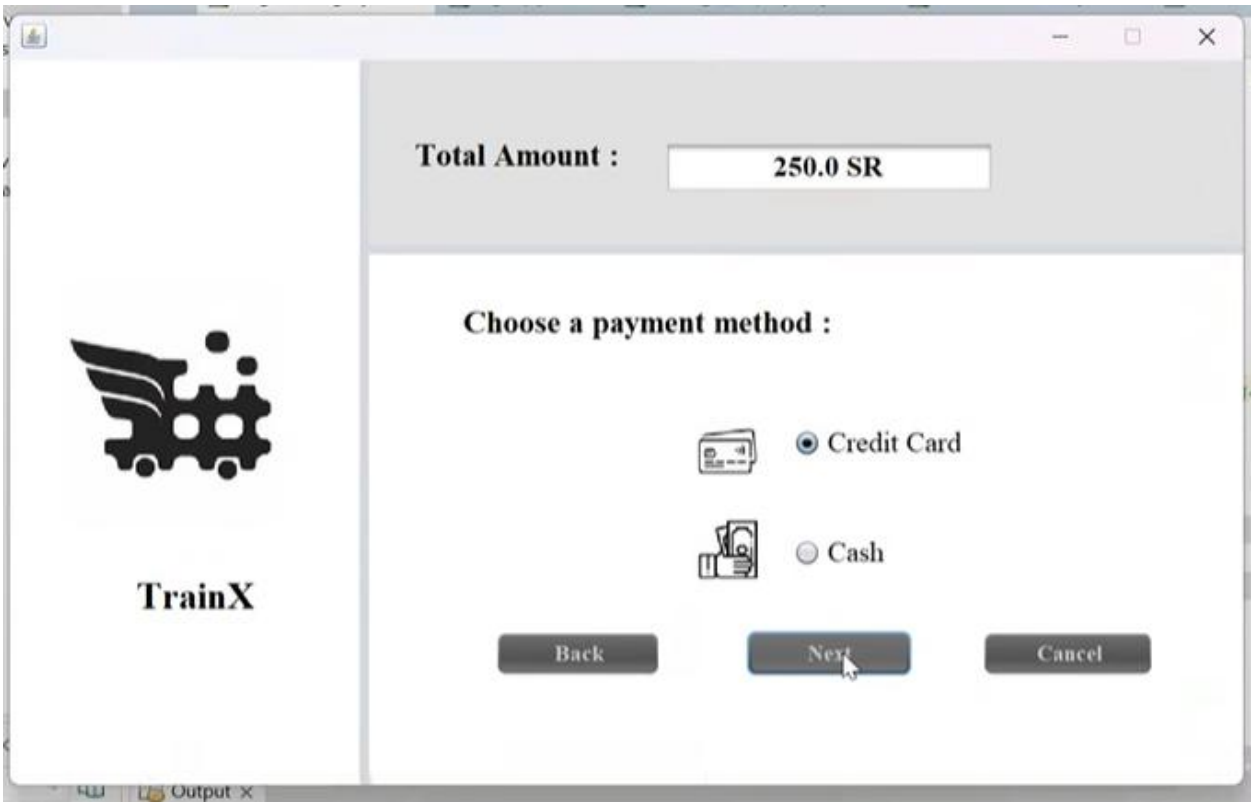
Confirm button will take the passenger to Check out.



The user can choose the payment method.

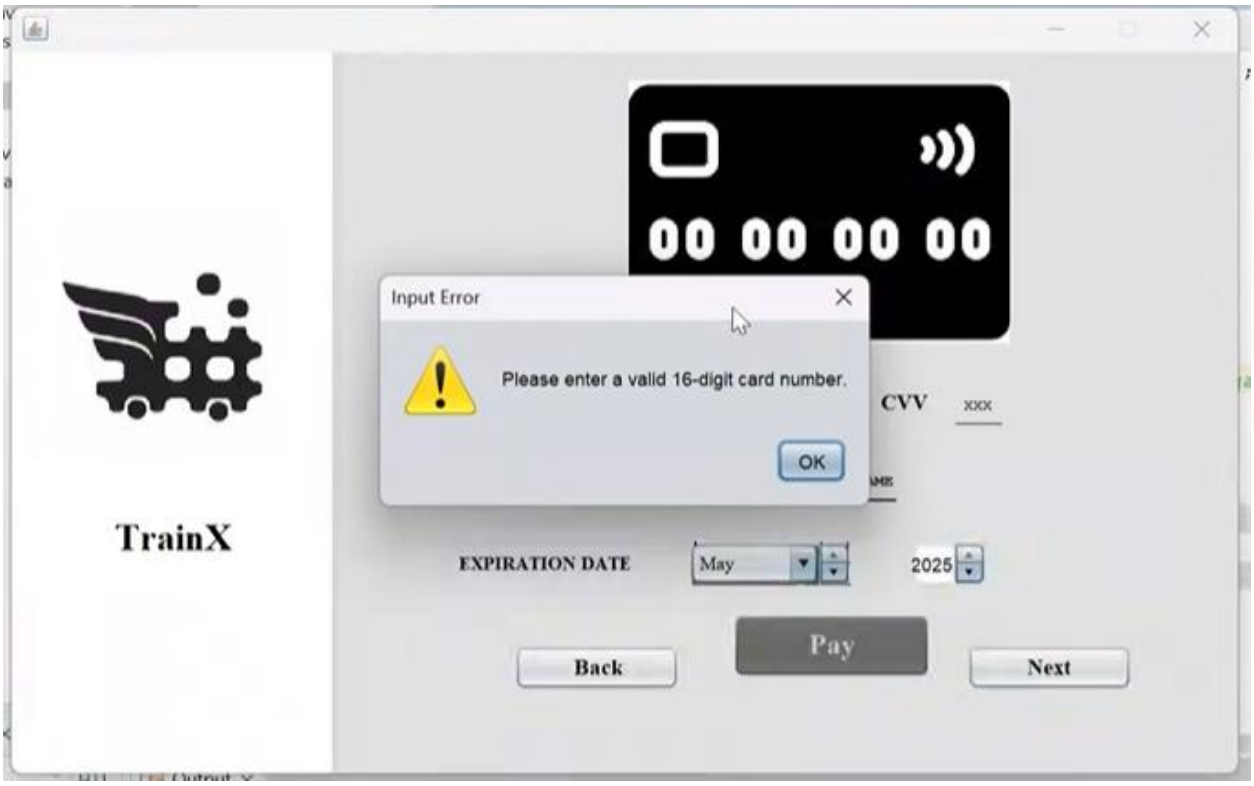


If the user choose Cash the program will inform them to go to the counter.

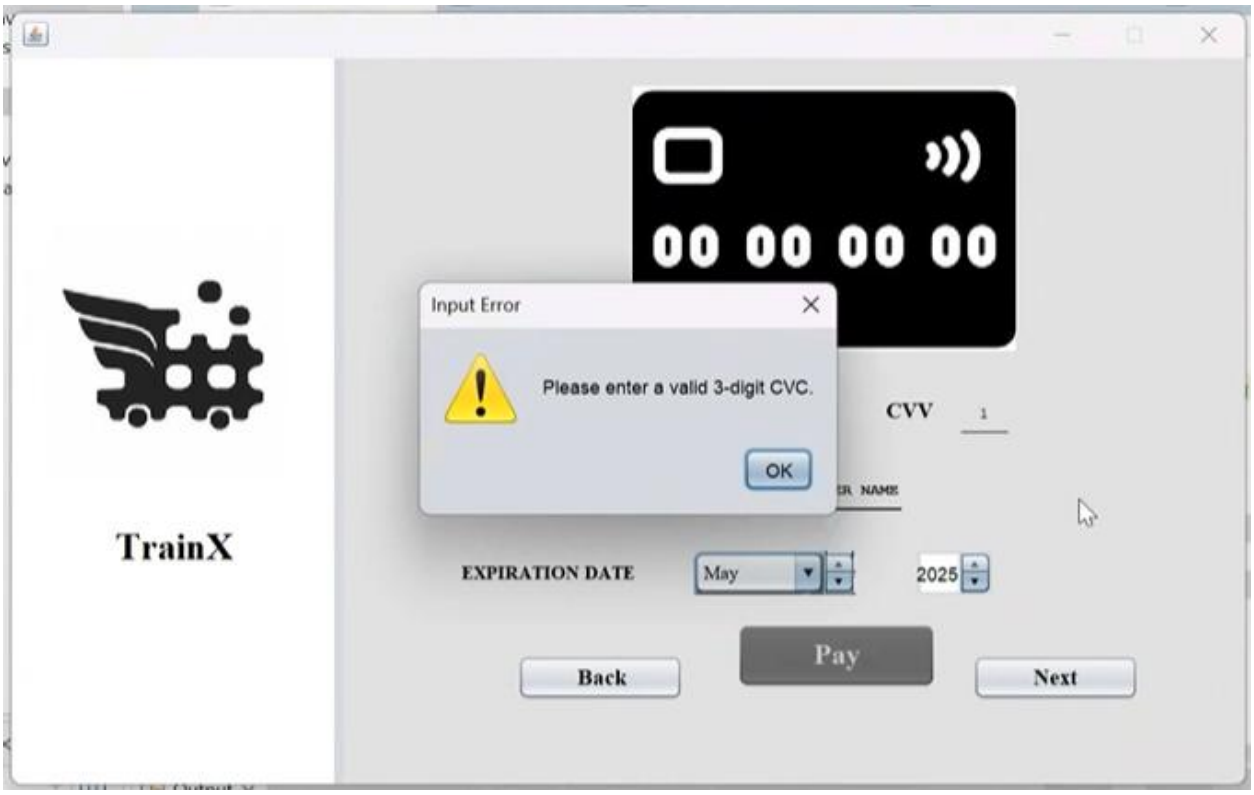


If the user choose Credit Card they can complete to pay with the application.

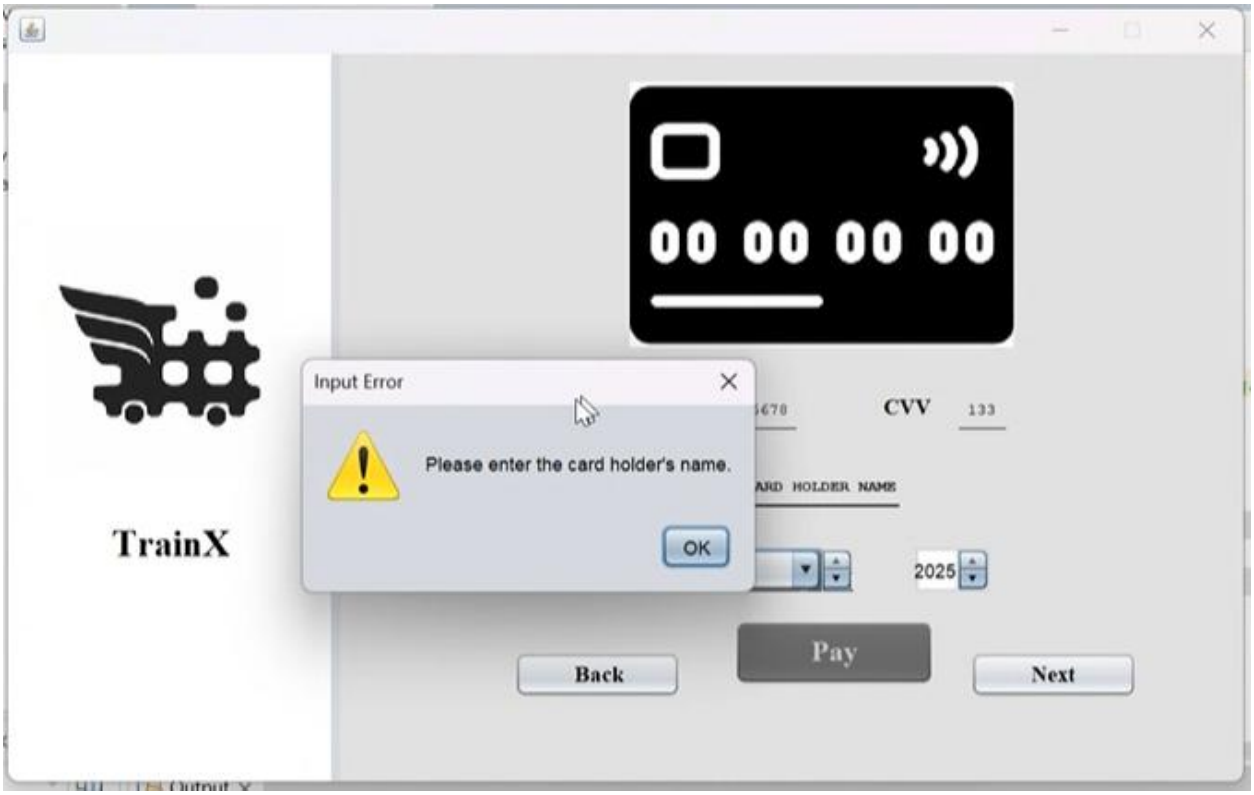
The Credit Card interface required the user card information



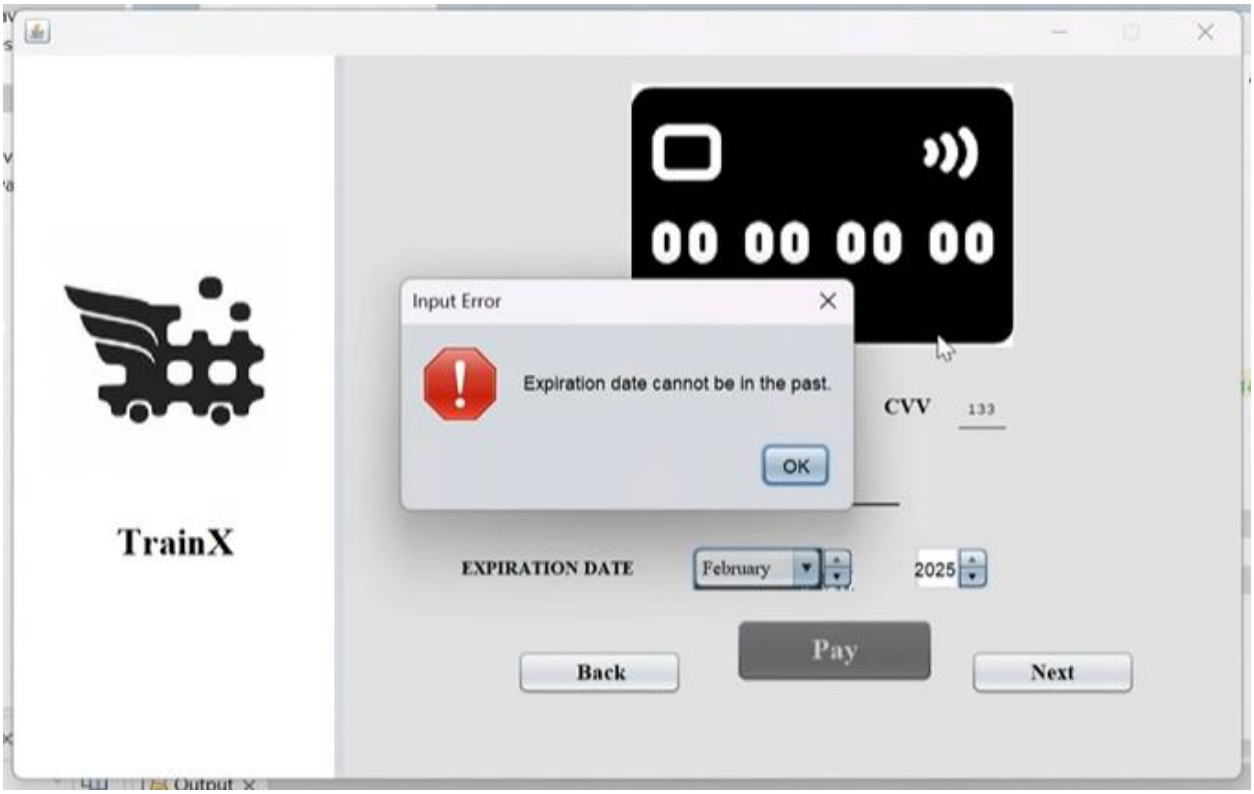
The user should enter 16 digit for the card number.



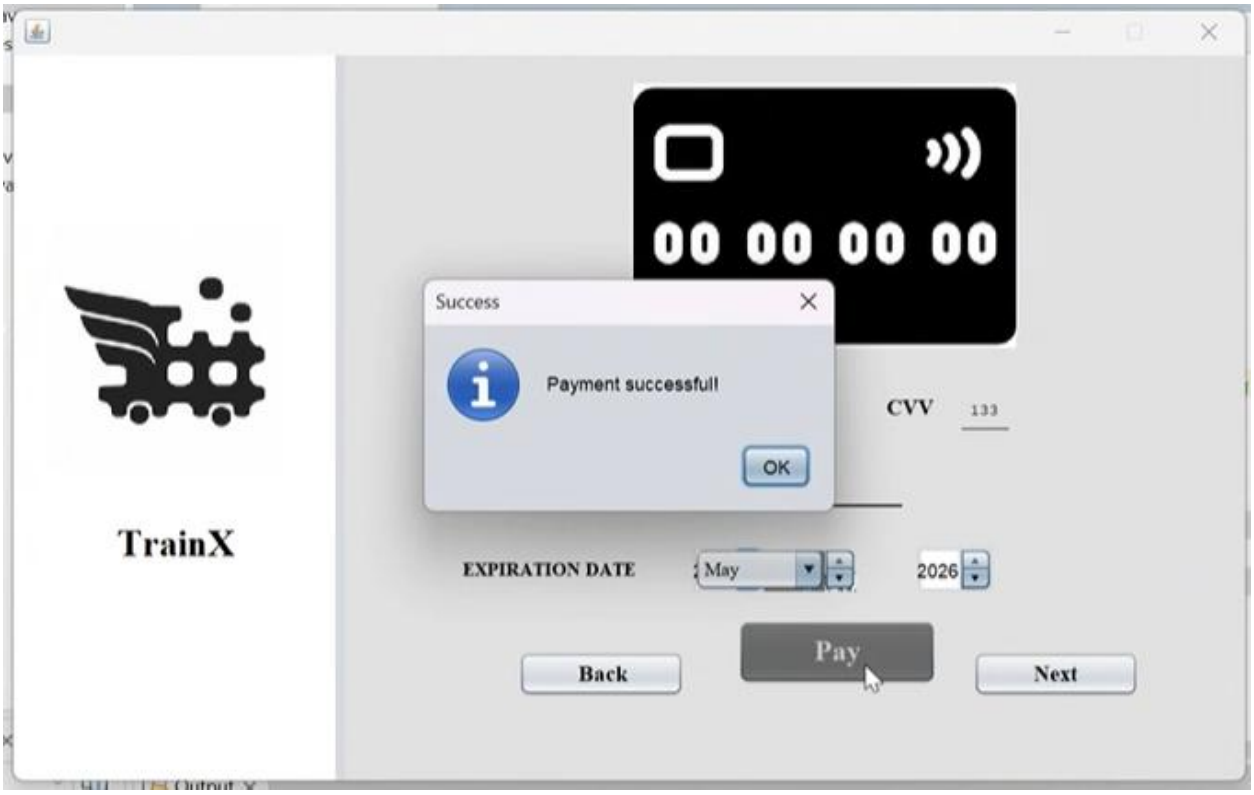
The user should enter 3 digit for the CVC.



The user should enter The Card holder's name.

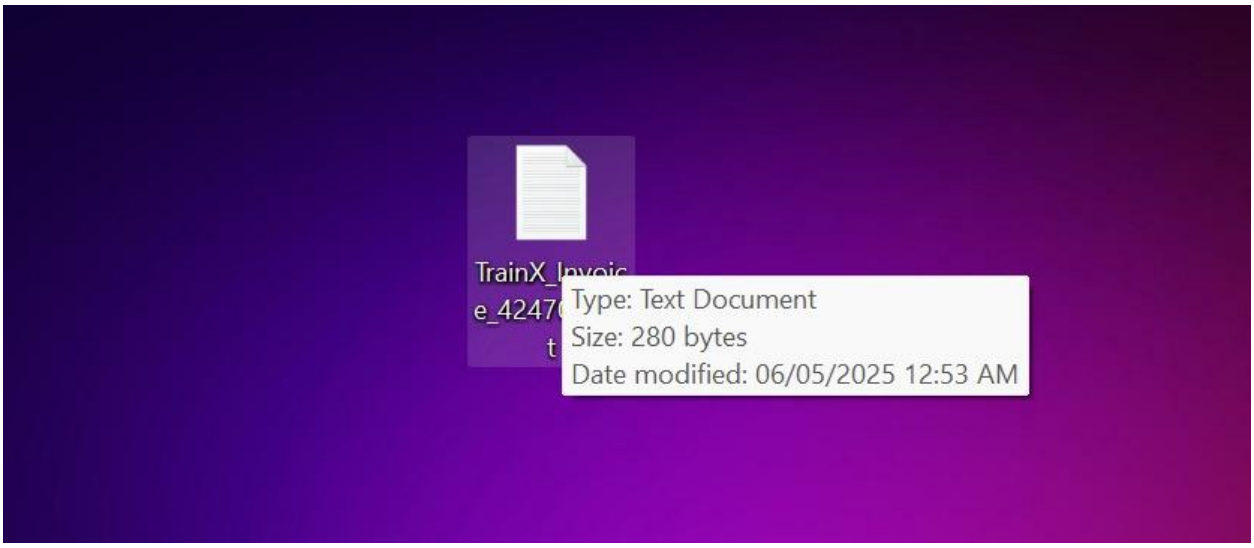


The user should enter a valid date for the expiration date.

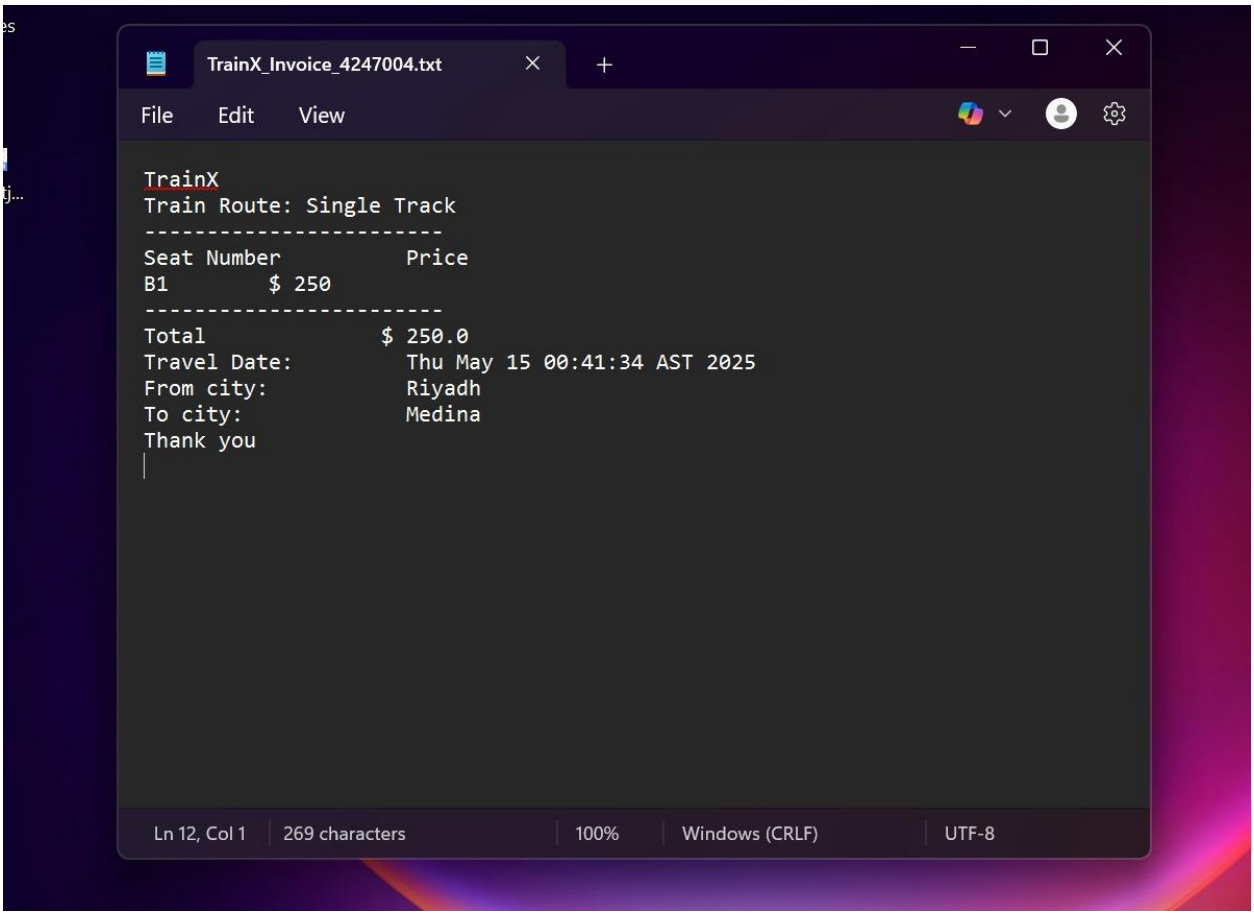


Then the payment Successful.

And inform the user that the invoice was saved in their desktop as a txt file.



This is the Invoice that saved befor.



And this is the Invoice Syntax that includes the trip details.